

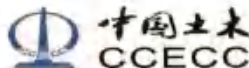
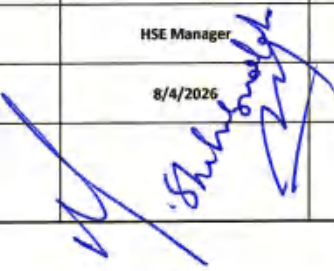
CLIENT 	CONSULTANT 	CONTRACTOR 												
METHOD STATEMENT SUBMITTAL														
PROJECT: R1181/2 Improvement of Al Safa Road														
FOR INFORMATION AND RECORDS	FOR APPROVAL	Reference No. : CCECC-R1181-2-CC-ST-MSRA-019 Rev. No. : 01 Date : 8-Apr-26												
To: Parsons Overseas Limited P.O.Box 9123 Dubai - U.A.E. From: China Civil Engineering Construction Corporation P.O.Box 393249 Dubai, U.A.E. Originator: Mohsen D Gazzaz Project Director														
Document Title:- METHOD STATEMENT AND RISK ASSESSMENT FOR PILING WORKS		RECEIVED 09 APR 2026 PARSONS R1181-2 Stamp for receipt (Consultant)												
Location of related works:- As per Approved Shop Drawings Date of Approval Required by: ASAP														
Other Remarks: <table style="width:100%; border-collapse: collapse;"> <tr><td style="width: 5%;">1</td><td>Method Statement (MS) Attached.</td></tr> <tr><td>2</td><td>Risk Assessment (RA) Attached.</td></tr> <tr><td>3</td><td>PTW Procedure Attached.</td></tr> <tr><td>4</td><td>Inspection & Test Plan (ITP) Attached.</td></tr> <tr><td>5</td><td>Checklist Attached.</td></tr> <tr><td>6</td><td>Emergency Contact Details Attached.</td></tr> </table>			1	Method Statement (MS) Attached.	2	Risk Assessment (RA) Attached.	3	PTW Procedure Attached.	4	Inspection & Test Plan (ITP) Attached.	5	Checklist Attached.	6	Emergency Contact Details Attached.
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3	PTW Procedure Attached.													
4	Inspection & Test Plan (ITP) Attached.													
5	Checklist Attached.													
6	Emergency Contact Details Attached.													
CLIENT/ENGINEERS RESPONSE (Please tick relevant box)														
C. s / Engineer's Comments: 1. To strictly adhere with the Engineer's comments as noted on the attached MSRA & Document Review sheets. 2. The length of the temporary casing & the drilling fluid to be used shall be determined based on actual site conditions & shall be submitted to the Engineer for approval prior to casing fabrication & commencement of piling activities. 3. A mock-up/trial bore is to be arrange on-site to demonstrate borehole stability, polymer performance and casing installation under actual site conditions. 4. A separate Method Statement shall be submitted for both working & non-working test piles for piling works.		Stamp for receipt: ☐ Approved ☒ Approved as Noted ☐ Revise & Resubmit ☐ No Approval Required												
For Client / Engineer Representative:														
Name: Project Director	Reference: 													
Signature:	Date:	16 APR 2026												

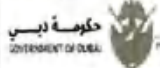
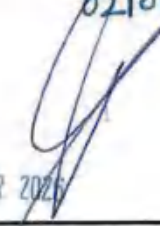
DISTRIBUTION		FILE NO:		
CCECC				
PROJECT: RTA (R118112) - IMPROVEMENT OF AL SAFA STREET				
DOCUMENT / CORRESPONDENCE DISTRIBUTION				
TO	INITIALS	ACTION	REVIEW	INFO
O.D				
D.O.D				
PD/PM	✓			✓
UPM				✓
PLANNING				
COMMERCIAL / QS				✓
HSE		✓		
TRAFFIC		✓		
DESIGN / TECHNICAL		✓		
QA/QC / MATERIAL		✓		
NOC				
CONSTRUCTION RR		✓		
CONSTRUCTION RO				✓
CONSTRUCTION UTIL				✓
SURVEYOR				
STORE				
HEAD OFFICE				
SUBCON				
OTHERS				

17/4/26

Client: 		Consultant: 		Contractor: 	
PROJECT: R1181/2 - IMPROVEMENT OF AL SAFA ROAD					
Comments Response Sheet (CRS)					
Ref. No. :	CCECC-R1181-2-CC-ST-MSRA-019 - Method Statement and Risk Assessment for Piling Works				
Rev. No. :	01				
Consultant : Parsons Comments /Client : RTA Comments					
S. No.	Consultant : Parsons Comments	CCECC Response to Rev.00 Comments	Remarks		
	Construction methodology for pier 18 piles to be added.	Noted & Complied (Please Refer Sec 4.12)			
1	If polymer slurry will not be used, the temporary steel casing shall be installed through loose or unstable soils and extended a minimum of 1.0 m into the hard strata.	Noted & Complied (Please Refer Sec 4.4)			
2	For cases involving a combination of temporary steel casing and drilling fluid (polymer), a separate method statement shall be submitted.	Noted & Complied (Please Refer Sec 4.4)			
3	The Contractor shall submit the piling sequence for each pile group, especially where a vibrator is used for installing or extracting the temporary casing.	Noted & Complied (Kindly refer Sec 4.9)			
4	Depending on the site conditions and surrounding utilities, the use of a vibrator for installing or extracting the temporary casing may not be permitted. NOC to be obtained from concerned authorities.	Noted. Vibrator usage will be subject to site conditions and surrounding utilities. Necessary NOCs will be obtained from concerned authorities prior to commencement.			
5	The Contractor has to explain that the minimum required concrete cover will not be disturbed during extraction of long casing using the piling rig.	Minimum cover to reinforcement (as per design requirement) shall be maintained.			
6	If damage is, or is likely to be, caused to utility services or adjacent structures, the Contractor shall propose appropriate shoring works, surveys, and movement monitoring to avoid such damage. A separate method statement shall be submitted.	Noted & Necessary shoring, surveys and movement monitoring will be implemented where required to protect utilities and adjacent structures. A separate method statement will be submitted for approval.			
7	Soil conditions at the pile toe shall be recorded, indicating the presence of rock (local or overall), soft or strong soil, or any abnormalities in the soil profile in the vicinity of the pile toe.	If any abnormality found; shall be recorded and informed to the Engineer.			
8	Any permanent deformation of the steel reinforcement cage due to storage or lifting conditions shall result in rejection of steel cage.	Noted. Any reinforcement cage with permanent deformation due to storage or lifting will be rejected.			
9	The Contractor shall explain the procedure for removing fine soil sedimentation when the recorded depth after drilling does not match the depth before concreting.	Once drilling completed up to required depth (+300 mm max.) and before lowering of rebar cage, cleaning the pile bottom shall be performed with cleaning bucket.			
10	In the event of rapid loss of drilling fluid or partial collapse of the borehole, lean concrete or cement mixed soil shall be used to backfill the borehole without delay.	Noted. In case of drilling fluid loss or borehole collapse, lean concrete or cement-soil mix will be used for immediate backfilling.			
11	Bored pile records shall comply with Project Specification Clause 8/6. The lifting plan shall be signed by the concerned Engineer.	Noted & Attached Bored Pile records as per Project Specs Cl 8/6.			
12	No concrete shall be placed until all driving within a radius of 4.5 m is completed, nor until all shells or casings for any one cluster have been completely driven. If this cannot be achieved, all driving within the above limits shall be discontinued until the concrete in the last pile cast has set for atleast two (2) days.	Noted & Shall maintain 4.5m distance of adjacent to the recently casted piles, if the pile casted within 24 hrs.			
13	Welding of steel reinforcement is not permitted.	Noted. Welding of steel reinforcement will not be carried out.			
14	To maintain the steel cage at the correct level, the cage shall be suspended from or restrained by the temporary casing until concreting is completed.	Noted. Steel cage shall be kept hanging on the temporary casing and the casing shall be removed after completion of concrete.			
15	All comments noted on the attached MSRA shall be incorporated.	Noted. All comments on the MSRA will be incorporated accordingly.			
16	If the Method Statement for piling works is not prepared by the piling contractor, it shall be formally reviewed, endorsed and signed by the piling subcontractor prior to submission for approval.	Noted. Method statement has been reviewed, signed, and sealed by the piling subcontractor and is attached for submission.			

Client: 		Consultant: 	Contractor: 
PROJECT: R1181/2 - IMPROVEMENT OF AL SAFA ROAD			
Comments Response Sheet (CRS)			
Ref. No. :	CCECC-R1181-2-CC-ST-MSRA-019 - Method Statement and Risk Assessment for Piling Works		
Rev. No. :	01		
Consultant : Parsons Comments /Client : RTA Comments			
S. No.	Consultant : Parsons Comments	CCECC Response to Rev.00 Comments	Remarks
	HSE Comments		
17	General : The contractor shall ensure safe execution of the works for Piling and associated activities by complying with the approved Project HSE Plan, CEMP, ERP, Logistic Plan. Compliance of all applicable local authorities' approvals and NOCs, TRA QHSS, MSTG, WZTMM, RTA Circulars and OM Code of Construction Safety Practice and contractual requirements.	Noted. To ensure safe execution of piling and associated activities in full compliance with the Project HSE Plan, CEMP, ECBCP / ERP, TMP, CWMP, approved permits and authority NOCs, RTA QHSS MSTG, RTA WZTM requirements, Dubai Municipality Code of Construction Safety Practice, and contractual requirements. Activity-specific controls have been strengthened for utility protection, working platform certification, open borehole safety, drilling, concreting, casing extraction, public protection, environmental controls, emergency response, welfare, and implementation of supporting sub-plans.	
18	NOC & Approvals: Sec 4.1 : Provide all authority approvals and utility clearance drawings.	Noted . The method statement has been revised to clearly confirm that no piling work shall commence without valid authority approvals, utility clearance drawings, approved shop drawings, approved TMP / ROW arrangements where applicable, and the required Internal PTWs. It is also clarified that all permits and NOCs shall remain valid for the entire activity duration and shall be renewed in advance where required.	
19	Utilities Protection: Sec 4.1 : Utility detection must be ensured with proper methodology and risk controls.	Noted. To include a dedicated utility protection requirement confirming that underground services shall be identified through approved utility drawings, NOCs, utility detection, trial trenching, and physical site marking prior to drilling. It is further stated that no mechanical excavation or drilling shall proceed near known or suspected utilities unless the safe clearance, supervision, and protection measures are in place, and any unidentified service shall trigger immediate stop work and authority notification.	
20	Working Platform: Sec: 4.3: The ground bearing capacity or platform stability certification must be confirmed.	Noted - The method statement has been revised to confirm that the piling working platform shall be designed / assessed, prepared, compacted, and certified as suitable for the imposed rig, crane, and concrete truck loads before plant mobilization. Daily inspections, post-rain inspections, and immediate rectification of soft spots, rutting, settlement, edge instability, or water logging have also been included.	
21	Temporary Casing: Sec 4.5: The safety measures for open boreholes must be detailed with all points.	Noted. This MS to include detailed safety measures for temporary casing installation and open borehole protection, including exclusion zones, lifting control, alignment and verticality checks, supervision by competent personnel, and strict prevention of personnel access near suspended casing sections. It is also clarified that open boreholes shall never be left unprotected and shall be rigidly barricaded, covered where practicable, illuminated for night work, and continuously monitored.	
22	Borehole Drilling: Sec4.6 : Key hazards missing (collapse, fall into borehole, entanglement). Add control measures.	Noted - To include the key drilling hazards previously omitted, including bore collapse, fall into borehole, entanglement with rotating parts, drilling tool failure, instability due to weak strata, and drilling fluid / slurry issues. Controls now include rigid barricading around bore locations, no loose clothing near rotating equipment, continuous bore stability monitoring, stop-work criteria for instability, and protection / covering of boreholes when drilling is paused.	
23	Concreting Works: Hazards like hose failure, overflow, and collapse are not addressed.	To address hose failure, pipe joint failure, concrete overflow, splashing, tremie instability, and uncontrolled discharge during concreting. It now confirms pre-pour inspection of pump lines, clamps, tremie joints, communication between pump operator and pour supervisor, continuous monitoring of tremie embedment, controlled pumping rate, overflow management, spill containment, and immediate stoppage if leakage, clamp movement, or abnormal pressure is observed.	
24	Casing Removal / Cutting: Sec 4.9 & 4.10: Hazards such as flying debris, noise, and falling materials are not included.	To include hazards and controls for casing extraction and pile head cutting, including flying debris, concrete splash, falling materials, noise, vibration, hand injury, and uncontrolled movement of extracted casing. It is now stated that extraction shall be carried out under continuous supervision while maintaining sufficient concrete head, and that pile cutting shall be done within a barricaded zone using suitable PPE including eye / face and hearing protection.	
25	Public Safety: General : Public Interface risks are not covered. Provide segregation and access control measures.	Noted - The method statement has been revised to specifically address public interface risks. It now confirms segregation of plant movement from pedestrian routes, rigid barricading of piling areas, controlled access points, traffic marshals / banksmen, protected walkways where applicable, warning signage, night lighting, and immediate reinstatement of any displaced barriers or public protection arrangements.	

Client: 		Consultant: 		Contractor: 	
PROJECT: R1181/2 - IMPROVEMENT OF AL SAFA ROAD					
Comments Response Sheet (CRS)					
Ref. No. :	CCECC-R1181-2-CC-ST-MSRA-019 - Method Statement and Risk Assessment for Piling Works				
Rev. No. :	01				
Consultant : Parsons Comments /Client : RTA Comments					
S. No.	Consultant : Parsons Comments	CCECC Response to Rev.00 Comments		Remarks	
26	Environmental: Environmental impacts such as slurry, waste management, noise are not detailed	Noted - The method statement has been revised to detail the environmental controls for piling works, including management of slurry, spoil, wastewater, concrete waste, noise, dust, and leaks / spills from plant and hoses. It is clarified that slurry / contaminated water shall be collected in designated lined pits / tanks, waste shall be removed by approved disposal routes, spill kits shall be available, discharge to ground / drainage is prohibited, and noise / dust suppression measures shall be implemented and monitored.			
27	Access/Egress: Sec 7.6: Not related to the actual activity. To be revised.	Noted & Complied - The access / egress section has been revised to make it activity-specific to piling operations rather than generic. It now defines designated access routes for piling rigs, cranes, transit mixers, pumps, and workers; segregation of pedestrian and plant routes; emergency escape routes; lighting requirements; route maintenance; control of soft ground / waterlogging; and continuous monitoring by site supervision.			
28	Emergency Response: Sec 7.12 & 7.15: Emergency plan/procedure is not specific to piling risks (collapse, flooding, equipment failure).	Noted & Complied. (Please refer MS Sec - 7.12) - The emergency response section has been revised to make it specific to piling risks, including bore collapse, worker fall into excavation / bore zone, utility strike, rig instability / overturning, hose burst, concrete spill, flooding / rain ingress, fire, equipment failure, and medical emergency. It now includes clear stop work / evacuation triggers, emergency communication flow, rescue equipment availability, standby pumps, isolation of affected zone, and authority / ambulance notification requirements.			
29	Welfare Facilities: Sec 7.5: Welfare arrangements are not detailed (rest shelter, first aid, drinking water, toilets etc.).	Noted & Complied. (Please refer MS Sec - 7.5) - The method statement has been revised to detail welfare arrangements, including shaded rest shelter, potable drinking water at work front, toilets / wash facilities, first aid availability, emergency transport, heat stress control, and cleaning / inspection of welfare facilities. It is also clarified that names / contact details of first aiders and emergency numbers shall be displayed and communicated to the workforce.			
30	Supporting Documents : All supporting documents such as Lifting Plan, ERP, TMP, CEMP, CWMP should be followed during the activity.	Noted. - The method statement has been revised to explicitly confirm implementation of all applicable supporting documents during piling works, including Lifting Plan, TMP, CEMP, CWMP, ERP / ECBCP, PTW procedure, and approved inspection / testing requirements. These supporting documents are now referenced as mandatory implementing documents for the respective activity stages, not merely as general reference documents.			
Sub-Contractor (Zemtas)		Contractor (CCECC)			
Name	Srinivasa Rao	Mirza Shahabuddin Baig	Syed Abdul Moid		
Entity / Function	Technical Engineer	HSE Manager	QA/QC Manager		
Date	8/4/2026	8/4/2026	8/4/2026		
Signature					

CLIENT 		CONSULTANT  PARSONS		CONTRACTOR  中国土木 CCECC	
METHOD STATEMENT SUBMITTAL					
PROJECT: R1181/2 Improvement of Al Safa Road					
FOR INFORMATION AND RECORDS		FOR APPROVAL		Reference No. : CCECC-R1181-2-CC-ST-MSRA-019 Rev. No. : 00 Date : 18-Mar-26	
To: Parsons Overseas Limited P.O.Box 9123 Dubai - U.A.E.		From: China Civil Engineering Construction Corporation P.O.Box 393249 Dubai, U.A.E.		 Originator: Mohsen D Gazzaz  Project Director	
Document Title:- METHOD STATEMENT AND RISK ASSESSMENT FOR PILING WORKS		 Stamp for receipt (Consultant)			
Location of related works:- As per Approved Shop Drawings Date of Approval Required by: ASAP					
Other Remarks: 1 Method Statement (MS) Attached. 2 Risk Assessment (RA) Attached. 3 Inspection & Test Plan (ITP) Attached. 4 Checklist Attached. 5 Emergency Contact Details Attached.					
CLIENT/ENGINEERS RESPONSE (Please tick relevant box)					
Clients / Engineer's Comments: <i>Risk Assessment approved as noted comments - comments sheet attached - Construction methodology for pier-18 piles to be added.</i> <i>- Refer to the Engineer's comments on the attached Document Review sheet, SH&E & MSRA</i>				30 MAR 2026 9/30 AM Stamp for receipt	
				☐ Approved ☐ Approved as Noted ☒ Revise & Resubmit ☒ No Approval Required	
For Client / Engineer Representative:					
Name: Signature:		Reference: Date: 02 APR 2026		02/04/26 	

DOCUMENT REVIEW SHEET

Project Name : R1181/2 – Improvements of Al Safa Road
Employer : Roads and Transport Authority (RTA)
Engineer : Parsons Overseas Limited
Contractor : China Civil Engineering Construction Corporation

Subject/ Title	Method Statement and Risk Assessment for Piling Works	Received Date	30/03/2026
Document Ref.No.	CCECC-R1181-2-CC-ST-MSRA-019	Returned Date	April 2, 2026
Revision No:	00	Final Approval Status	C-Revise and Resubmit

Sl. No.	Engineer's Review Comments	Reviewer Initials	Contractor's Response	Status
1	If polymer slurry will not be used, the temporary steel casing shall be installed through loose or unstable soils and extended a minimum of 1.0 m into the hard strata.	RS		
2	For cases involving a combination of temporary steel casing and drilling fluid (polymer), a separate method statement shall be submitted.	RS		
3	The Contractor shall submit the piling sequence for each pile group, especially where a vibrator is used for installing or extracting the temporary casing.	RS		
4	Depending on the site conditions and surrounding utilities, the use of a vibrator for installing or extracting the temporary casing may not be permitted. NOC to be obtained from concerned authorities.	RS		
5	The Contractor has to explain that the minimum required concrete cover will not be disturbed during extraction of the long casing using the piling rig.	RS		

Any review, comment or acceptance by Employer/ Engineer under this contract with respect to any of contractor's documents shall not in any way be, or deemed to be, an approval of any works or an acceptance that the contractor's documents meet the requirements of the contract, as contractor has the sole responsibility for performing the works in accordance with the requirements of this contract.


 02/04/26

6	If damage is, or is likely to be, caused to utility services or adjacent structures, the Contractor shall propose appropriate shoring works, surveys, and movement monitoring to avoid such damage. A separate method statement shall be submitted.	RS		
7	Soil conditions at the pile toe shall be recorded, indicating the presence of rock (local or overall), soft or strong soil, or any abnormalities in the soil profile in the vicinity of the pile toe.	RS		
8	Any permanent deformation of the steel reinforcement cage due to storage or lifting conditions shall result in rejection of the steel cage.	RS		
9	The Contractor shall explain the procedure for removing fine soil sedimentation when the recorded depth after drilling does not match the depth before concreting.	RS		
10	In the event of rapid loss of drilling-fluid or partial collapse of the borehole, lean concrete or cement-mixed soil shall be used to backfill the borehole without delay.	RS		
11	Bored pile records shall comply with Project Specification Clause 8/6. The lifting plan shall be signed by the concerned Engineer.	RS		
12	No concrete shall be placed until all driving within a radius of 4.50 m is completed, nor until all shells or casings for any one cluster have been completely driven. If this cannot be achieved, all driving within the above limits shall be discontinued until the concrete in the last pile cast has set for at least two (2) days.	RS		
13	Welding of steel reinforcement is not permitted.	RS		

Any review, comment or acceptance by Employer/ Engineer under this contract with respect to any of contractor's documents shall not in any way be, or deemed to be, an approval of any works or an acceptance that the contractor's documents meet the requirements of the contract, as contractor has the sole responsibility for performing the works in accordance with the requirements of this contract.

BMF
02/04/26

14	To maintain the steel cage at the correct level, the cage shall be suspended from or restrained by the temporary casing until concreting is completed.	RS		
15	All comments noted on the attached MSRA shall be incorporated.	RS		
16	If the Method Statement for piling works is not prepared by the piling subcontractor, it shall be formally reviewed, endorsed, and signed by the piling subcontractor prior to submission for approval	RS		

Any review, comment or acceptance by Employer/ Engineer under this contract with respect to any of contractor's documents shall not in any way be, or deemed to be, an approval of any works or an acceptance that the contractor's documents meet the requirements of the contract, as contractor has the sole responsibility for performing the works in accordance with the requirements of this contract.

Stamp
02/04/26

SH&E Review Comments

Project Name:	Improvement of Al Safa Road	Project #	R1181/2	Date	24.03.2026
Contractor:	China Civil Engineering Construction Corporation	Reviewer	Ghulam Moinuddin HSE Engineer (Parsons)	Sign:	
Document Title:	Method Statement & Risk Assessment for Piling Works	Document #	CCECC-R1181/2-CC-ST-MSRA-019	Rev	00
Status	B = Approved with Comments Status A = Approved, Status B = Approved with Comments, Status C = Revise and Resubmit, Status D = Rejected				

The submitted MSRA is Approved with comments, subject to incorporation of the attached review comments. The Contractor shall ensure all comments are addressed and implemented prior to commencement of the activity.

Title / Topic	Ref	Parsons's Review / Comments	Contractor's Inputs	Ref	Parsons Status
General		The contractor shall ensure safe execution of the works for Piling and associated activities by complying with the approved Project HSE Plan, CEMP, ERP, Logistic Plan. Compliance of all applicable local authorities' approvals and NOCs, TRA QHSS MSTG, WZTMM, RTA Circulars and DM Code of Construction Safety Practice and contractual requirements.			
NOC & Approvals	Sec 4.1	Provide all authority approvals and utility clearance drawings.			C
Utilities Protection	Sec 4.1	Utility detection must be ensured with proper methodology and risk controls.			C
Working Platform	Sec 4.3	The ground bearing capacity or platform stability certification must be confirmed.			C
Temporary Casing	Sec 4.5	The safety measures for open boreholes must be detailed with all points.			C
Borehole Drilling	Sec 4.6	Key hazards missing (collapse, fall into borehole, entanglement). Add control measures.			C
Concreting Works		Hazards like hose failure, overflow, and collapse are not addressed.			C
Casing Removal / Cutting	Sec 4.9-4.10	Hazards such as flying debris, noise, and falling materials are not included.			C
Public Safety	General	Public interface risks are not covered. Provide segregation and access control measures.			C
Environmental		Environmental impacts such as slurry, waste management, noise are not detailed			C
Access/Egress	7.6	Not related to the actual activity. To be revised.			C
Emergency Response	Sec 7.12-7.15	Emergency plan/procedure is not specific to piling risks (collapse, flooding, equipment failure).			C
Welfare Facilities	Sec 7.5	Welfare arrangements are not detailed (rest shelter, first aid, drinking water, toilets etc.).			C

Supporting Documents	General	All supporting documents such as Lifting Plan, ERP, TMP, CEMP, CWMP should be followed during the activity.			C
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STRUCTURE WORKS

METHOD STATEMENT AND RISK ASSESSMENT FOR PILING WORKS



HISTORY

Referential		Submission :	☐ For Information Only ☒ For Approval	
Rev	Date	Modifications Content	Writer	Function
00	4-Mar-26	First Issue	Jeevanantham. R	QA/QC Engineer
01	8-Apr-26	First Issue	Jeevanantham. R	QA/QC Engineer

CCECC REVIEW AND VERIFIED BY :

	REVIEWER 1	REVIEWER 2	REVIEWER 3	REVIEWER 4	REVIEWER 5	VERIFIED BY
Name	Srinivasa Rao	Syed Abdul Mold	Mirza Shahabuddin Baig	Ahmed Abdalla	Ahmed Elkadi	Mohsen D Gazzaz
Entity / Function	Technical Engineer	QA/QC Manager	HSE Manager	Technical Manager	Deputy. Project Manager	Project Director
Date	8-Apr-26	8-Apr-26	8-Apr-26	8-Apr-26	8-Apr-26	8-Apr-26
Signature						



CONTENTS

1	Purpose and Scope	4
1.1	Purpose	4
1.2	Scope	4
2	Definitions and References	4
2.1	Definitions	4
2.2	Document References	5
3	Process Resources	6
3.1	Personnel Roles and Responsibilities	6
3.1.1	Project Director	6
3.1.2	Deputy Project Manager	6
3.1.3	Construction Manager	6
3.1.4	Technical Manager	6
3.1.5	QAQC Manager	7
3.1.6	QA/QC Engineer	7
3.1.7	Material Engineer / Lab Technician	7
3.1.8	HSE Manager / HSE Officer	7
3.1.9	Traffic Management Team (TTM Manager / Engineer)	8
3.1.10	Site Engineer	8
3.1.11	Foreman / Chargehand	8
3.1.12	Chief Surveyor / Surveyor	8
3.1.13	Workers, Operators & Drivers	9

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 1 of 39

3.1.14	Subcontractor (MNG Zemtaz Soil & Foundation Engineering Construction Co. Inc)	9
3.2	Tools and Equipment	10
3.3	Material	11
3.4	Manpower	11
3.5	Transportation	11
4	Procedure	12
4.1	Prior Activities	12
4.2	Preparatory Works	13
4.3	Surveying and Setting up at the pile location	13
4.4	Installation of Temporary Casing & Positioning	13
4.5	Drilling of Pile Borehole	15
4.6	Reinforced Cage Installation	17
4.7	Casting Concrete	18
4.8	Extraction of Temporary Casing	19
4.9	Sequence of Piling Works:	19
4.9	Cutting off the Pile Heads	19
4.10	Pile testing	20
4.11	Cutting off the Pile Heads	20
4.12	Construction of Piling Works Near the Underpass Abutment Walls	20
4.13	Preliminary Test Pile	22
5	Activity Hold Points	22
6	Inspection and Records	23

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 2 of 39

7	Safety and Environmental Risks	23
7.1	Hazard & Risk Assessment	23
7.2	Tools and Equipment	23
7.3	Accident / Incident	23
7.4	Protective and Safety Equipment	24
7.5	Welfare Arrangements	24
7.6	Access / Egress to Construction Area	25
7.7	List of Training	26
7.8	First Aid	27
7.9	Lighting	27
7.10	Working Outdoors	27
7.11	Moving Around The Premises	27
7.12	Emergency Response Procedures	28
7.13	Rescue Arrangements	31
7.14	Emergency Contact Details	32
8	Appendices	33
8.1	Appendix A: Risk Assessment	
8.2	Appendix B: PTW Procedure	
8.2	Appendix C: Inspection & Test Plan	
8.3	Appendix D: Checklist	
8.4	Appendix E: Emergency Contact Details	

METHOD STATEMENT FOR PILING WORKS

1. Purpose and Scope

1.1 Purpose

The purpose of this method statement is to lay down the systematic step by step procedures for carrying out the Structural Piling Works within the Scope of CCECC.

This is to verify and to ensure that works are carried out in accordance with the approved shop drawings and project specifications.

1.2 Scope

The Scope of this method statement is to carry out the Piling works as per the Approved Shop Drawings and Project Specification.

All the works shall be supervised by the Site Engineer and Engineers Representative for the **Project: R1181-2 - IMPROVEMENT OF AL SAFA ROAD.**

2. Definitions and References

2.1 Definitions

Client	Roads & Transport Authority (RTA)
Consultant	Parsons Overseas Limited (POL)
Contractor	China Civil Engineering Construction Corporation – Dubai (CCECC)
Subcontractor	M/s. MNG Zemtaz Soil & Foundation Eng Construction Co. Inc.
DEWA	Dubai Electricity & Water Authority
DM	Dubai Municipality

Title:	Method Statement for Piling Works	R1181-2 - IMPROVEMENT OF AL SAFA ROAD
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.: 01
		Page No.: Page 4 of 39

2.2 Document References

ISO 9001:2015	Quality Management Systems - Requirements
ISO 45001:2018	Occupational Health and Safety Management System - Requirements
ISO 14001:2015	Environmental Management System - Requirements
Technical Specifications	Project Specifications
PAS 128:2014	Specification for underground utility detection, verification, and location.
Dubai Municipality Control Stations	As and where applicable
Dubai Code of Construction Safety Practice	As and where applicable
RTA / DM Requirements	RTA Work Zone Traffic Management Manual , DMCOCS, MSTG & RTA Circulars.
Federal Law No. (24) Of 1999	Concerning the Protection, Development and Amendments of the Environment (amended in Law No. (11) Of 2006)
DM NRCS Technical Guideline	Translocation, Handling, and Restoration of Wildlife (DM-ED-GU-01-THRWTG).
DEWA	Water Technical Standards and Specifications & DEWA Handbook on EHV Over Headline & Underground cable protective
BS 1377 Part 9: 1990	Methods of Test for Civil Engineering Purpose
BS 5930: 1999	Codes of practices for site investigation
CCECC-R1181-2-CC-DS-001	Project Quality Plan (PQP)
CCECC-R1181-2-CC-DS-002	Health and Safety Construction Management Plan (HSCMP)
CCECC-R1181-2-CC-DS-003	Construction Environment Management Plan (CEMP)
CCECC-R1181-2-CC-DS-005	Construction Waste Management Plan (CWMP)
CCECC-R1181-2-CC-DS-007	Construction Lifting Management Plan (CLMP)
CCECC-R1181-2-CC-DS-006	Traffic Management Plan (TMP) (Where as applicable)
CCECC-R1181-2-CC-DS-010	Emergency / Contingency & Business Continuity Plan (ECBCP)
CCECC-R1181-2-CC-DS-015	Project Logistics Management Plan (PLMP)

Title:	Method Statement for Piling Works	R1181-2 - IMPROVEMENT OF AL SAFA ROAD
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.: 01
		Page No.: Page 5 of 39

3. Process Resources

3.1 Personnel Roles and Responsibilities

3.1.1 Project Director

- Provide overall direction and resources to execute the works safely and as per contract requirements.
- Ensure project HSE Plan and PQP are implemented and supported by management.
- Review and close critical issues / high-risk non-conformances escalated by site team.
- Liaise with Client/Consultant/Authorities on major matters when required.

3.1.2 Deputy Project Manager

- Manage daily site delivery and ensure works follow approved drawings, method statement and MSRA.
- Ensure manpower, plant, materials and competent supervision are available before work starts.
- Ensure inspections/hold points and approvals are arranged as per ITP and RFA requirements.
- Monitor progress and address constraints affecting safe execution.

3.1.3 Construction Manager

- Plan and control the activity sequence and ensure safe work arrangements are in place before execution.
- Coordinate site engineer, QAQC and HSE for required inspections, hold points and closeouts.
- Ensure competent supervision is present during critical activities and stop unsafe work when required.
- Ensure subcontractor performance and compliance with method statement and HSE requirements.

3.1.4 Technical Manager

- Manage shop drawings, technical submittals and as-built documentation.
- Coordinate with service authorities for approvals.
- Raise and resolve RFIs and technical discrepancies.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 6 of 39

- Maintain submittal logs and document control.

3.1.5 QAQC Manager

- Implement and monitor the Project Quality Plan.
- Review and approve ITPs, Method Statements and Material Submittals.
- Ensure inspections are conducted as per approved procedures.
- Issue and close NCRs and coordinate with Consultant on quality matters.

3.1.6 QAQC Engineer

- Verify materials and workmanship comply with approved drawings, specifications and ITP.
- Coordinate RFAs/NMRs and ensure required inspections are completed before proceeding to next stage
- Ensure formwork/rebar and pre-pour checklists are completed and documented before concrete pour.
- Maintain quality records including test results, approvals and inspection reports.

3.1.7 Material Engineer / Lab Technician

- Ensure only approved materials are used and maintain material approvals and delivery records
- Coordinate required sampling/testing and maintain test logs and reports
- Verify calibration certificates for testing equipment and maintain calibration records
- Review test results and inform QAQC/Site Engineer of any non-conformance for corrective action

3.1.8 HSE Manager / HSE Officer

- Verify risk controls in MSRA are implemented before and during the activity
- Conduct pre-start checks, toolbox talk coordination and routine safety inspections
- Ensure PTW/high-risk controls, exclusion zones, access/egress and housekeeping are maintained
- Stop unsafe work and ensure corrective actions and closeout evidence are completed
- Ensure emergency arrangements (first aid, ambulance contact, emergency access) are in place

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 7 of 39

3.1.9 Traffic Management Team (TTM Manager / Engineer)

- Implement approved TMP/TTM arrangements including signage, cones, barriers and lighting
- Maintain safe access/egress and pedestrian management as per approved plan
- Provide trained flagmen/marshals where required and monitor compliance during works
- Coordinate with authorities/permits for any lane closure/diversion requirements

3.1.10 Site Engineer

- Control execution of the activity as per approved drawings, method statement and MSRA
- Provide clear instruction to supervisors/foremen and ensure compliance is verified at the work front
- Arrange required inspections/hold points and ensure approvals are obtained before proceeding
- Ensure plant positioning, work area layout and exclusion zones are established and maintained
- Coordinate with QAQC/HSE for records, closeouts and corrective actions

3.1.11 Foreman / Chargehand

- Direct workers, allocate tasks and ensure work is done as per Site Engineer instruction and method statement
- Ensure tools/equipment are used correctly and unsafe practices are not permitted
- Maintain housekeeping, safe access/egress and barricading/exclusion zones at the work area
- Report any unsafe condition, near miss or damage immediately to Site Engineer and HSE.

3.1.12 Chief Surveyor / Surveyor

- Carry out setting-out and level checks as per approved drawings and benchmarks
- Verify dimensions/levels prior to concrete pour and issue survey records as required
- Maintain calibrated instruments and survey control points

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 8 of 39

3.1.13 Worker, Operator & Driver

- Follow instructions, method statement and site rules at all times
- Use only authorized equipment and operate plant only if competent/licensed
- Conduct pre-use checks and report defects immediately
- Use mandatory and task-specific PPE and comply with barricades/exclusion zones

3.1.14 Subcontractor (MNG Zemtas Soil & Foundation Engineering Construction Co. Inc)

The subcontractor will carry out Piling & associated works in coordination with Main contractor.

- Follow Project Specification and Contract Drawings requirements.
- Follow all the Quality and Safety requirements in the Project.
- Have a safe site, plant, structures and ways of working.
- Mobilize and handle all required resources.
- Make sure plant, structures and substances are used, handled and stored safely.
- Provide facilities for the welfare of workers, such as drinking water and toilets.
- Provide the necessary information, training and supervision to protect people from risk.
- Monitor health and workplace conditions.
- Coordinate with CCECC personnel for submitting inspection requests and arranging inspections.
- Assist for inspections, Keep CCECC personnel about day to program.
- Coordinate with CCECC personnel for piling concrete.
- Ensure all the required submittals, method statement, PQs are submitted in time.
- Prepare and submit pile bore logs with all required information in time.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 9 of 39

3.2 Tools and Equipment

- Temporary Casing
- Tremie pipes
- Augers & Drilling buckets
- Hydraulic Drill Rig with accessories
- Crawler crane for lifting works (lowering of steel, concrete casting, etc.)
- Vibro Hammer (if required)
- Water Tanks
- Polymer Tanks
- Delivery Pump (if required)
- Air Compressor (7 to 15 bar)
- Wheel loader (Shovel) or Backhoe loader
- Mobile Crane – If necessary
- Hand Tools
- Concrete Pumps (Including spare pumps)
- Concrete Transit Mixers

Note: -

- 1 The above listed equipment or additional equipment shall be mobilized subject to the requirement as we need necessary.
- 2 Calibration Certificates will be available as applicable for each instrument used on site.
- 3 All Machineries, Equipment's shall be "Third Party Inspected and Certified.
- 4 All Machineries, Equipment's shall be labeled with QR Code and Maintain the Color Code.
- 5 All Operators shall be "Third Party Certified".

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 10 of 39

3.3 Material



- Ready Mix Concrete as per approved Mix design
- Reinforcement Steel
- Tie wires or Binding wires
- Cover Blocks
- Polymer (depending on soil condition)
- Concrete Spacers
- Sonic tubes
- Temporary sign boards, barricades etc.

3.4 Manpower

- Construction Manager
- Site Engineer
- Safety Officer
- Site Supervisor / Foremen
- Foreman, Chargehand
- Technicians
- Helper / Laborers
- Operators,
- Drivers

3.5 Transportation

- For all Vehicles / equipment's to be used on site safety equipment such as fire extinguisher, first aid kit, vehicle emergency equipment and communication devices will be providing. The operators shall be in possession of necessary licenses and third part certificates of competence of the equipment's to be used.
- Site speed limit should be 20 km/h and should not exceed in any case.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 11 of 39

4. Procedure

4.1 Prior Activities

- All necessary NOC to obtain earlier than work resumption and duration to be renewed in advance in case of activity is prolonged due to any unavoidable reason.
- Obtain approval for all related documentation and TTM drawings, including the Permit to work, the relevant method statement, and the construction NOC from Engineer and Relevant authority prior to start the activity on site.
- Work Start Notification shall be submitted to Engineer prior to commencement of work at site
- All details shall be included in the shop drawings prior to submission to the Engineer for approval.
- Review geotechnical soil investigation report, handing over the indorsed/approved drawings for work to the personnel to work at site.
- Identify and mark underground utilities & ensure proper site drainage.
- All shop drawings shall be obtaining approval from the Consultant and related authority for further execution and inspection.
- Mobilize piling rigs and supporting equipment, Ensure equipment inspection and valid certification.
- During execution, if any deviations observed that should be included in the shop drawing as revision, and that shall be submitted to Engineer.
- Setup necessary traffic safety arrangements according to the approved traffic management plan.
- Conduct the necessary safety induction to all Team members involved in the Activity, Approved equipment shall be checked, ready to use and available on site.
- A representative of the Engineer's organization will witness, observe and record the works as per approved shop drawings, Method Statement and ensure that all the operations comply with the contract specifications.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 12 of 39

4.2 Preparatory Works

- All Works are carried out in accordance with the Contract specification, approved shop drawing, and as per the material submittals and manufacturer recommendations.
- Proper Access/Egress to construction area will be done.
- Working platform must have a well-compatible soil or solid base to ensure stability of the equipment.
- The platform level of the pile will be confirmed as per the shop drawing.
- Prior to drilling, NOC from all concerned authorities will be obtained.
- Shop drawings shall be approved before commencement of the work.

4.3 Surveying and Setting up at the pile location:

- The pile locations shall be set out using Total Station by the contractor's surveyor.
- Accuracy of pile positioning is critical as deviation will affect structural alignment.
- Allowable tolerance shall be as per project specification (typically $\pm 75\text{mm}$).
- Pile point shall be checked and approved by the Engineer before commencement.
- Final confirmation shall be made after temporary casing installation.

4.4 Installation of Temporary Casing & Positioning:

- The pile bores will be stabilized with a temporary casing in the upper layers, temporary casing of single wall construction will support the upper, unstable sub-strata. The length of the casing will be determined from actual conditions encountered on site for the Engineer approval. *Approved polymer shall be used*
- Single wall (S/W) temporary casing of an average length of 10m will normally be installed by a rotational turntable driven by the boring rig.
- Temporary casing shall be installed at pile location using rotary rig or vibro hammer.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 13 of 39



- Dimension and quality of casing shall be adequate to withstand all possible stress.
- Casing shall be of uniform cross section throughout and shall be free from any internal projection and stuck concrete that can prevent proper formation of pile.
- If casing is damaged during construction, it shall be removed, and joints of casing shall be reasonably watertight
- Casing alignment and verticality shall be checked using spirit level or plumb line.
- The equipment governing the alignment of the bored pile shall be made vertical to a tolerance of **within finished pile 1:75**.
- Proper alignment ensures full pile capacity and structural safety.
- Pile shall be installed from the existing ground level or approved working platform level.
- Pre-drilling down to required depth will be made if the stratum is so hard to drive the casing.
- The surveyor will mark the center of pile as per approved setting out drawing. From the center point of the pile (marked by a steel peg) four reference pegs are set 90 degrees to each other at a distance equal to 1 meter from the outside of the temporary casing to be installed to ensure the center of the pile.
- The verticality of the casing will be checked with a spirit level at every 2– 3m during the installation of casing.
- Verticality of the borehole will be achieved by properly setting up the rotary rig on the borehole location and ensuring the verticality of the casing which will serve as a guide for the drill string.
- The verticality of the pile will be continuously checked from the automatic level control of rig by checking the verticality of the Kelly-bar of the rig during drilling
- Once the Casing is installed, the coordinates will be **checked by the Consultant Surveyor and Engineer**.
- Verticality control during driving of steel casing and **it shall also be confirmed with spirit level**.



Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 14 of 39



- Temporary Steel casing shall extend not less than 0.5m above the working platform and where drilling fluid or ground water is present, not less than 1.0m to ensure proper containment of slurry /concrete and safe working conditions.



4.5 Drilling of Pile Borehole:

- Adequate safety measures shall be implemented prior to execution.
- The area around each pile location shall be leveled using JCB/Bobcat to provide stable working platforms for piling rigs and concrete trucks.
- After installation of temporary casing down to hard strata, Drilling will be performed (as per project specification and construction drawings).
- Drilling will continue by using appropriate drilling tools, completely until reaching the designed pile toe level by the rig's depth measuring with a meter tape with a dead weight at the end.
- Augers and/or buckets attached to piling rig will be used to drill boreholes of designed diameter. If hard strata are encountered special core barrels and heavy-duty rock augers with tungsten carbide teeth will be used to drill through them.
- The stability of borehole shafts will be realized by driving temporarily casing down to the hard strata. Drilling will continue without any drill fluid since we are using casing down to the hard strata or down to the weak rock (or even weathered rock)

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 15 of 39



to stabilize the drilled shaft.

- If shaft stability cannot be achieved with casing at deeper levels (in some cases the hard strata is deeper than max 15m), water as drilling fluid will be added into borehole to provide the stability of drilled shaft. If the soil is very weak/loose at deeper levels wherein the shaft cannot be stabilized by water, then polymer slurry will be added as per the project specifications, the level of fluid will be kept constant ~~level shall by 1.0m above the level of external standing ground water and all the to be performed for drilling fluid on site and will be~~ ^{test} monitored during the whole drilling process.
- Piles shall not be bored so close to other piles less than 4.5m of pile center to center which have recently been cast and which contain workable or unset concrete that a flow of concrete could be induced from or can cause damage to any of the piles. In such a case minimum 48hours will be waited for the concrete to set.
- Drilling penetration rate and drilling debris will be kept under observation for any change in the sub-soil conditions & verify the soil report.
- Pile lengths will be as per approved drawings / schedules or as otherwise instructed by the Engineer.
- Variations out with these drawings or schedules should be instructed in writing by the Engineer.
- When reaching the final depth, loose and re-moulded material and debris will be removed with the drilling or cleaning bucket to confirm a firm base.
- When the pile shaft boring has been completed, the drilled borehole shall be inspected and approved by the Engineer prior to commencement of reinforcement installation and concrete pouring. Checking borehole depth will be conducted by using an end weighted measuring tape to verify the depth given in the drawing.
- Final pile depths will be recorded and agreed prior to the placement of the reinforcement cage and prior to pouring concrete the depth shall be verified.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 16 of 39



Operator and rigger are in charge during boring

4.6 Reinforced Cage Installation:

- The reinforcement steel shall be uncoated of type-high yield deformed Bars in accordance with the latest copy of BS 4449:2005, with exception that the steel shall be Grade 500 as stipulated in General Specification -Volume 2-Part 1 section 4.
- Reinforcement cages shall be fabricated as per approved shop drawings.
- Spacer blocks shall be provided to maintain required concrete cover.
- Cages shall be lifted using crane and carefully lowered into boreholes.
- Cage levels shall be maintained by temporary support/hooks on casing.
- The diameter of bars and spacings shall be as per approved shop drawing.
- The rebars above Pile cut off level will be provided with plastic sleeves for debonding in case of working piles.
- Before lowering the cage, it should be inspected and approved by the Engineer. The cages shall be inspected by Engineer.
- Sonic **pipes (25% of all piles)** if applicable, shall be joined together and installed in

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 17 of 39

the cage. End caps shall be fixed at bottom and top of pipes. The number of pipes shall be as per approved shop drawing. The Sonic pipes shall be filled with water before casting the pile.

4.7 Casting Concrete:

- Tremie pipes shall be assembled and lowered to pile base level. The mouth of Tremie pipe at top will be connected to concrete pump and pumping of concrete will start. The approved grade and mix design will be used.
- Concrete received at site will be checked for slump and temperature. Cubes will be cast for compressive strength.
- It should be ensured that the bottom of tremie pipe shall remain at a minimum of **2.0 meters embedded** in fresh concrete during casting.
- **Concrete grade for the piles to be used in C45/20, except for abutment piles which are C50/20.**
- Concrete shall be placed continuously without interruption & **Concrete slump shall be checked as per project specs and mix design.**
- Pouring shall continue until all contaminated concrete comes out of the bore hole and a clean concrete flow starts coming out.
- During the concreting period, drilling fluid if any or ground water flushed from boreholes will be collected to the tanks or excavated pits with the help of Submersible pumps, and these tanks/pits should be emptied regularly for the next piling operation.



Concrete sample testing at site.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 18 of 39

4.8 Extraction of Temporary Casing:

- Upon completion of concreting, temporary casing shall be extracted while the concrete within them remains sufficiently workable to ensure that the concrete is not lifted. During **extraction of steel casing by rotary system** the motion of the casing shall be maintained in an axial direction relative to the pile.
- When the casing is being extracted, a sufficient quantity of concrete shall be maintained within it to ensure that pressure from external water, drilling fluid or soil is exceeded and that the pile is neither reduced in section nor contaminated
- The concrete level within a temporary casing shall be topped up where necessary during the course of casing extraction in such a way that the base of the casing is always below the concrete surface until the casting of the pile has been completed.

4.9 Sequence of Piling Works:



- A Shop drawings shall be submitted prior to the commencement of the work.
- Mark pile locations (Pile 1, 2, 3) and verify levels, Prepare and compact working platform.
- Position piling rig at Pile 1 and check alignment.
- Bore, clean, install reinforcement cage, and cast Pile 1, Shift rig to Pile 3 (skip Pile 2) **alternate piles concrete will be done** as per approved shop drawing.
- The distance between two piles shall be maintained at 4.5 m.
- Bore, clean, install reinforcement cage, and cast Pile 3 and Keep Pile 2 undisturbed.
- Move rig to Pile 2 and complete boring & casting.
- Record all data and check level, alignment, and quality.

Pile Casting & Boring Sequence (3 Adjacent Pile)

Step 1:



Step 2:



Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 19 of 39

4.10 Pile testing:

- Caliper logging test of Pile Bore shall be done as per Project specifications PS 8/10.16 (The Piles bigger than 1000mm).
- Pile Integrity tests shall be done according to project specifications on 8/17.
- Cross hole sonic test will be carried out for the piles as per approved drawings by the Engineer
- Static Load tests will be carried out for both working and non-working piles as per approved shop drawings by the Engineer.

Note: Prior to commencing any test, related test method statements will be submitted to the Engineer for approval.

4.11 Cutting off the Pile Heads:

- After concrete gains sufficient strength as per satisfactory cube test result, pile head shall be trimmed to required cut-off level.
- Cutting shall be done using pneumatic breakers carefully to avoid damage to reinforcement.
- Exposed reinforcement shall be cleaned and prepared for pile cap connection.
- Works shall comply with Project Specification Section 8/16.1 (or relevant clause).
- The 20cm above cut off level shall be break with electrical hammers.

4.12 Construction of Piling Works Near the Underpass Abutment Walls



Sequence of Work

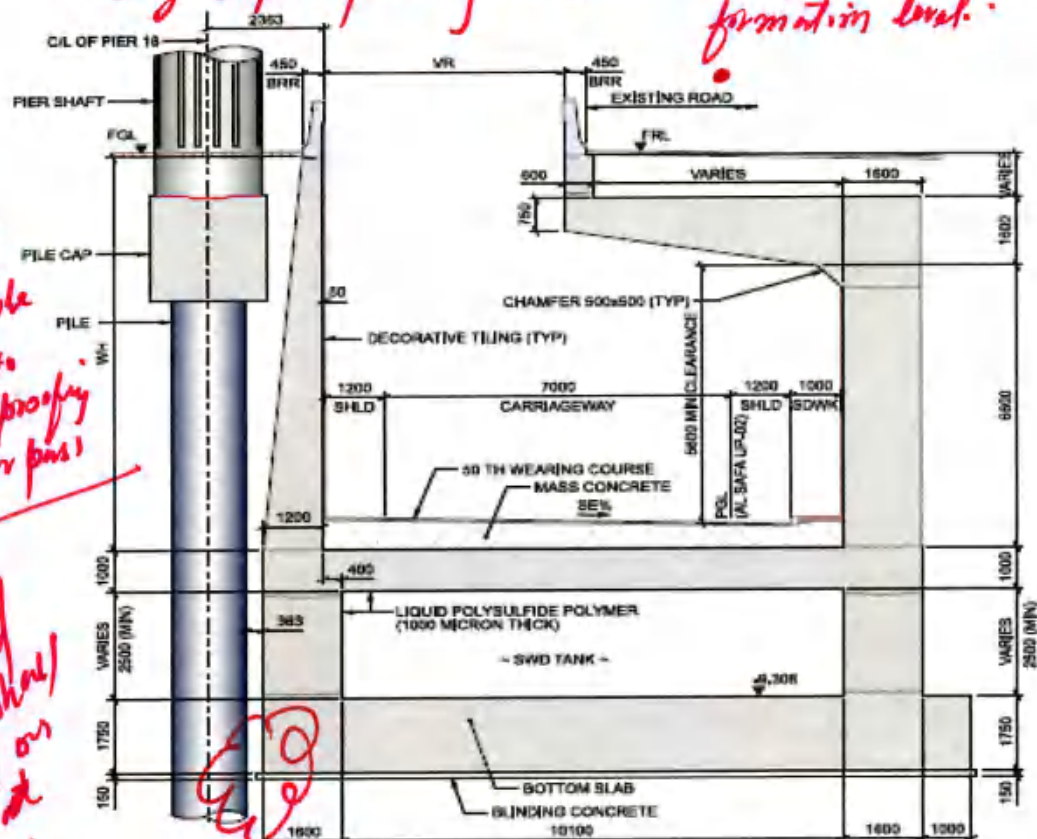
- Construction of Underpass Bass Slab as per Approved Method Statement Ref. No. CCECC-R1181-2-CC-ST-MSRA-028.
- Construction of Underpass Abutment Walls as per Approved Method Statement Ref. No. CCECC-R1181-2-CC-ST-MSRA-034.
- Application of Waterproofing System as per Approved Method Statement Ref. No.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 20 of 39

CCECC-R1181-2-CC-ST-MSRA-018/01.

- Backfilling to be done as per Approved Method Statement, Approved Shop Drawings and Project Specification.
- Piling Works near the Underpass Abutment Walls shall be done as per Approved Piling Design Approved Shop Drawing and Approved Method Statement.

• long temporary casing shall be used, shall reach below the formation level.



CROSS SECTION - UNIT (SAFA-UP1-11) AT PIER 18
FROM STA. 0+354 TO STA. 0+365

The following piles are doing above mentioned method

- Safa 1 - Bridge 5 - P18 & P19
- Safa 1 - Bridge 6 - P20 & Abutment - 2
- Safa 1 - Bridge 1 - Abutment 1 & Pier 1.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 21 of 39

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4.13 Preliminary Test Pile

Submit a separate MSRA for PTP covering all details..

- A Preliminary Test Pile is a pile installed before commencement of bulk piling works to verify the geotechnical design assumptions and confirm the ultimate load carrying capacity of the proposed pile system.
- It is typically constructed at a designated test location away from permanent structure (unless otherwise specified by engineer).

The Preliminary Test Pile main objectives are:

- Confirm soil bearing strata depth, verify soil parameters used in design.
- Assess skin friction, confirm pile structural behavior under high load, Check settlement characteristics.
- Evaluating creep performance, confirm working load and factors of safety.

Identify need for modification in:

- Pile diameter
- Pile length
- Reinforcement design

Note : Preliminary test piles are tested **2.0 times the Working Load** as per Project Specification PS 8/18.1.

5. Activity Hold Points

As reference for HSE, general activities related to work are:

- Permit to work
- Risk Assessment
- Induction and pre-start completed
- Plant and equipment safety checks completed.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 22 of 39

6. Inspection and Records

All inspection / check list and document records shall be based on field notes taken during activities and shall be recorded

HSE records shall include:

- Toolbox Talk Record
- Equipment inspection check-sheet (As applicable)
- Competency of operators, flagman and all site team involved in above mentioned method statement to be verified by the CCECC.

7. Safety and Environmental Risks

Hazards and risk assessment are carried out and attached in this Method Statement in the form of a Risk Assessment. These are reviewed by the HSE Manager / HSE Officer to ensure that all the hazards have been identified and adequate control measures are in place.

All personnel shall wear the appropriate PPE.

7.1 Hazard & Risk Assessment

A Risk Assessment will be completed by the section performing the work together with the HSE Department. This document will identify hazards and develop strategies to mitigate the risks. All personnel carrying out the work will be competent and experienced. Where applicable, certificates of competence shall be provided.

Refer to Attachment-A.

7.2 Tools and Equipment

All tools and equipment for the task shall be in a safe condition and fit for its purpose. Where applicable, registration papers and certificates are to be provided.

At the start of each shift, equipment shall be visually checked for mechanical and structural soundness. All equipment shall be carefully examined and all hoses and couplings checked for leaks and general conditions.

7.3 Accident / Incident

In the event of an incident, response will be as per the procedure. Prestart and Toolbox meetings will be carried out to reinforce safety requirements prior to commencement of the works.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 23 of 39

7.4 Protective and Safety Equipment

Mandatory Personnel protective equipment to be worn by all staff and operatives on this site to carry out the tasks of this method statement are:

- Safety Helmet
- Safety Boots
- High Visibility vest
- Safety goggles
- Safety Gloves as and when required
- Dust mask as and when required
- Ear protection as and when required

7.5 Welfare Arrangements

- All welfare and medical arrangements shall comply with Dubai Municipality requirements and the Project HSE Plan.
- Adequate welfare facilities shall be provided and maintained in hygienic conditions, including drinking water, toilets/wash areas and rest/shade area as applicable to site conditions.
- Welfare facilities shall be located reasonably close to work areas (where practicable) and access routes shall be kept safe, clean and well-lit for day/night use.
- Housekeeping and cleaning schedules shall be implemented, and records shall be maintained to ensure welfare facilities remain clean, serviceable and free from foul smell, leakage and waste accumulation.
- Potable drinking water shall be provided at site and at work fronts; clean cups/water dispensers shall be maintained and refilled as required.
- Availability of a first aid room/clinic (where applicable) shall be ensured as per project requirements.
- Trained and certified first aiders shall be available during all shifts, and their names/contact numbers shall be displayed at site notice boards, work fronts and site office.
- Emergency medical response arrangements (ambulance service/nearest clinic or hospital) shall be established; emergency contact numbers and site location details shall be displayed and communicated to all supervisors and workers.
- Emergency transport/standby vehicle shall be arranged where applicable for prompt transfer of injured persons to the designated medical facility.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 24 of 39

- All injuries/illness cases shall be recorded and reported as per company and project incident reporting requirements, and follow-up actions shall be implemented.

7.6 Access / Egress to Construction Area



- Clearly marked and designated access/egress routes should be planned. These paths should lead to / from the ground level in a manner that minimizes hazards and is easily recognizable.
- Roads must allow safe movement and turning of rigs, cranes, and trucks.
- Strengthen soft ground to prevent sinking or accidents & Minimize risk by avoiding interaction between workers and moving vehicles.
- Restrict and monitor site access through proper controlled systems & ensure only approved personnel and vehicles enter the site.
- Maintaining visibility, especially for night operations, clearly indicate escape routes and assembly areas.
- Identify and rectify any damage or unsafe conditions.
- Prevent slippery or unstable road conditions & Improve visibility and reduce health hazards.
- Avoid accidents caused by debris & ensure safe entry and exit at all times.
- Conduct daily supervision and toolbox talks Reinforce safety awareness and site rules regularly.
- Conduct soil investigation before piling & provide proper shoring or casing for boreholes.
- Maintain safe distance between piles and excavation edges.
- Avoid over-excavation and unsupported soil, Restrict heavy equipment movement near edges, Monitor ground stability continuously.
- Provide barricading around open boreholes, Ensure proper sequencing of piling activities.
- Stop working during signs of soil movement/cracks, always Supervise by competent engineer.
- Provide proper site drainage system & Install dewatering pumps where required.
- Monitor groundwater level regularly, Protect boreholes from water entry.
- Stop working during heavy rain conditions, keep emergency pumps available on site, ensure discharged water is safely diverted.
- Prevent water accumulation near piling rigs, Maintain access roads free from water logging.
- Conduct pre-start inspection of all equipment, Ensure valid third-party certification for machinery

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 25 of 39

- Use only trained and certified operators, do not overload cranes or rigs
- Perform regular maintenance and servicing, Keep emergency repair tools available
- Stop equipment immediately if there is abnormal noise/vibration.
- Maintain safe distance from operating equipment, Provide standby equipment if critical.

7.7 List of Training

1	Emergency Response & Evacuation Procedure Training	
2	Power Tools & Equipment Safety Training	
3	Permit to Work (PTW) System Awareness Training	
4	Manual Handling & Ergonomics Training	
5	First Aid Awareness Training	
6	Equipment Handling Awareness Training	
7	Environmental Protection & Spill Response Training	
8	Fire Safety & Hot Works Awareness Training	
9	Materials Storage Safety Training	
10	Confined Space Awareness Training	
11	Working at Height Safety Training	
12	Ladder Safety Training	
13	Personal Protective Equipment (PPE) Training	
14	Night work Training	
15	Housekeeping & Waste Management Training	
16	Heat Stress Management Training	
17	Slip trip fall Awareness Training	
18	Accident, incident/near miss reporting	
19	Traffic Management & Site Access Safety Training	
20	Noise & DUST Control Training	

Note: The required training shall be provided at the time of site activity, based on the specific task, risk level, and site conditions.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 26 of 39

7.8 First Aid

- A fully stocked first aid kit shall always be present on site, accessible and regularly inspected.
- A minimum of one certified First Aider shall be present during all work activities.
- Emergency transportation shall be available in case medical evacuation is needed.
- A designated First Aid Station shall be established if the site conditions and manpower size require.

7.9 Lighting

- Good light will be provided– use natural light where possible and avoided glare
- A good level of local lighting at Workstation where necessary will provided.
- Suitable forms of emergency lighting
- Well-lit corridors and Access.
- Well-lit outside areas – for pedestrians and to help with work activities such as loading/unloading at night.

7.10 Working Outdoors

- For work outdoors we consider things such as the weather, temperature (both hot and cold) and sun exposure. Work will be as per DM CODE OF SAFETY & APPROVED SUMMER PLAN.

7.11 Moving Around the Premises

- Safe passage for pedestrians and vehicles – separate routes will be provided.
- Personal Protective Equipment is mandatory
- Levelling, even ground surfaces without holes or broken boards
- Handrails and ramps where necessary
- Set up properly secured fall arrest systems (like lanyards connected to fixed anchor points).
- Ensure no work is carried out during adverse weather conditions like high winds or rain, which increase the risk of falls.
- Create designated areas for safe movement and keep others away from high-risk zones.
- Safely constructed Access, Surfaces which are not slippery.
- Earth Moving Activity will be ensured as per the approved Earth Moving Equipment Plan.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 27 of 39

7.12 Emergency Response Procedures



This emergency procedure shall apply to all piling-related activities including:

- Site mobilization
- Working platform preparation
- Survey and setting out
- Utility identification / trial trenching
- Rig assembly and movement
- Temporary casing installation
- Borehole drilling
- Reinforcement cage lifting and lowering
- Tremie pipe installation
- Concrete pouring
- Casing extraction
- Pile head cutting
- Testing and associated activities.

Emergency Scenarios Specific to Piling Works

Include the following emergencies clearly:

- Borehole collapse / instability
- Person falling into borehole / excavation / restricted work zone
- Rig overturning / rig instability / working platform failure
- Temporary casing collapse / misalignment / dropping
- Utility strike (power cable, water line, telecom, gas if applicable)
- Fire or explosion from plant, fuel, or electrical fault
- Concrete hose burst / tremie failure / concrete overflow
- Lifting failure / dropped load / suspended load incident
- Flooding of borehole / excavation due to rain or utility rupture
- Major plant or mechanical failure during drilling or lifting
- Medical emergency / heat stress / collapse of worker
- Traffic accident or public interface incident near the piling area.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 28 of 39

A. Borehole Collapse / Ground Instability

- Stop drilling immediately.
- Remove all personnel from the danger zone.
- Barricade the borehole / collapse area.
- Suspend nearby piling, excavation, and plant movement.
- Assess adjacent ground condition and working platform stability.
- Do not allow any rescue attempt without competent supervision and safe access assessment.
- Inform Engineer / Project Management immediately.
- Resume only after technical review and approval.

B. Fall into Borehole / Excavation Area

- Stop all machinery and isolate the area.
- Do not allow uncontrolled entry or rescue attempt.
- Use trained rescue team / standby rescue arrangement.
- Call ambulance immediately.
- Provide retrieval only using safe method and proper rescue equipment.
- Maintain crowd control and preserve access for emergency responders.

C. Utility Strike

- Stop work immediately.
- Evacuate the immediate area.
- Do not touch damaged cable / pipe / wet area / exposed utility.
- Isolate plant and ignition sources.
- Inform concerned authority immediately:
- Barricade and secure area until authority clearance is received.
- No re-start until written / official clearance is obtained.

D. Rig Instability / Rig Overturning / Platform Failure

- Stop rig operation immediately.
- Move all personnel away from rig radius and fall zone.
- Shut down the machine if safe to do so.
- Isolate the working platform area.
- Prevent approach by other plant.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 29 of 39

- Assess platform settlement / edge failure / soft ground.
- Recovery shall only be by approved engineering method and lifting / recovery plan.

E. Temporary Casing Failure / Dropped Casing

- Stop drilling / lifting immediately.
- Establish exclusion zone.
- Remove personnel from fall radius.
- Recover only under approved lifting supervision.
- Inspect borehole stability before recommencing.

F. Fire / Fuel Leak / Electrical Fire

- Stop work and switch off engine / generator / power source if safe.
- Raise alarm and isolate fuel source.
- Use appropriate fire extinguisher only if fire is small and manageable.
- Evacuate if fire escalates.
- Call Civil Defence / emergency services immediately.
- No smoking / ignition source permitted nearby.

G. Concrete Hose Burst / Tremie Failure / Concrete Spill

- Stop pumping immediately.
- Isolate pump and pressure source.
- Keep personnel away from hose line.
- Control spill and prevent discharge to drainage.
- Check for injuries from splash / pressure / whip.
- Re-inspect clamps, tremie joints, and pump line before restart.

H. Flooding / Water Ingress / Rain Emergency

- Stop work immediately during unsafe rain or flooding.
- Switch off electrical equipment in the affected area.
- Remove personnel and plant from soft / flooded area.
- Protect boreholes from overflow and collapse.
- Activate dewatering / standby pumps.
- Inspect working platform and access roads before resuming work.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 30 of 39

I. Medical Emergency / Heat Stress

- Stop work and inform first aider / nurse immediately.
- Shift casualty to safe shaded area if possible.
- Provide first aid without delay.
- Call ambulance for serious condition.
- Maintain clear route for emergency vehicle access.
- Record and report incident as per project procedure.

J. Public / Traffic Incident

- Stop adjacent plant movement immediately.
- Secure public access route.
- Inform traffic team / marshal / TTM engineer.
- Call ambulance / police if injury or road incident is involved.
- Reinstate traffic control devices and safe pedestrian segregation immediately.

7.13 Rescue Arrangements

Rescue arrangements shall be specific to irrigation construction activities and shall not rely on generic or unrelated methods.

Excavation / Trench Rescue:

- No entry into collapsed trench without assessment
- Trained rescue team shall respond
- Use of ladders, shoring support, and safe access
- Emergency services to be contacted immediately for major collapse

Confined Space (if applicable – manholes):

- Tripod, harness, lifeline system shall be used
- Standby man present at all times
- Gas monitoring and ventilation ensured

Flooding / Dewatering Failure:

- Immediate evacuation from excavation
- Pumps activated / backup system engaged
- Area secured to prevent re-entry

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 31 of 39

Traffic-related Incident:

- Area secured using traffic control measures
- Casualty moved only if safe
- Coordination with Traffic Police and ambulance

Lifting Operation Incident:

- Exclusion zone maintained
- No attempt to move suspended load manually
- Crane isolated and area secured

Medical Emergency / Heat Stress:

- Immediate first aid, Transfer to shaded/cooling area
- Ambulance called if required

Rescue Equipment Availability:

- Stretchers, first aid kits, Rescue ropes, harness
- Spill kits, Fire extinguishers,

Rescue Team & Responsibility:

- Trained personnel assigned for emergency response
- Roles clearly defined and communicated

All rescue arrangements shall be briefed during toolbox talks and verified through drills.

Mock Drills: Regular emergency mock drills shall be conducted to ensure all personnel are familiar with rescue procedures and equipment use.

7.14 Emergency Contact Details

In case of Emergency Contact Telephones, No.:

- Police : 999
- Ambulance : 998
- Civil Defence : 997
- HSE Coordinator : 0561487012
- HSE Manager : 0559331082

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 32 of 39

8. Appendices

8.1 Appendix A: Risk Assessment

8.2 Appendix B: PTW Procedure

8.3 Appendix C: Inspection & Test Plan

8.4 Appendix D: Checklist

8.5 Appendix E: Emergency Contact Details

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 33 of 39

Appendix A: Risk Assessment

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 34 of 39



HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

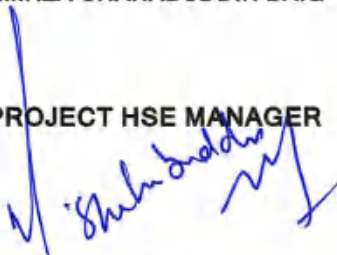
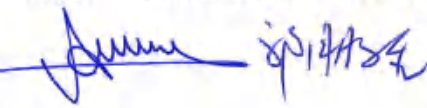
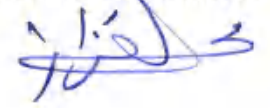
RISK ASSESSMENT FOR PILING WORKS

PROJECT: R1181-2 - IMPROVEMENT OF AL SAFA ROAD

1. DOCUMENT CONTROL.

Contractor / Subcontractor:	China Civil Engineering Construction Corporation - Dubai Branch	Project Location:	Al Safa Street (Financial Cent Road to Al Wasl Road)	Last Revision Date:	Not Applicable
Method Statement Ref. No:	CCECC-R1181-2-CC-ST-MSRA-019	Risk Assessment Ref. No:	CCECC-R1181-2-HSE-RA-19	Next Review Date:	15-10-2026
Assessment Date:	08-04-2026	Activity Type:	☒ Routine ☐ Non-Routine	Revision No:	01

This Risk Assessment shall be reviewed and updated upon any change in methodology, design, equipment, regulatory requirements, traffic stage/diversion, identification of utility conflicts, or following any accident or incident.

REVIEW & AUTHORIZATION.	RISK ASSESSED BY ENGINEER + SITE MANAGER +CM	REVIEWED BY HSEM	REVIEWED BY DPM	AUTHORIZED BY PD
NAME:	SRINIVASA RAO	MIRZA SHAHABUDDIN BAIG	AHMED ELKADI / GUO PENGCHENG	MOHSEN D GAZZAZ
DESIGNATION:	TECHNICAL ENGINEER	PROJECT HSE MANAGER	DEPUTY PROJECT MANAGER	PROJECT DIRECTOR
SIGNATURE:				
DATE:	08-04-26	08-04-26	08-04-26	08-04-26

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

2. SCOPE OF WORK:



- The **Execution of Bored Cast In-Situ Piling Works** including mobilization of piling equipment, setting out, drilling of boreholes, installation of casing and reinforcement cage, tremie concrete pouring, casing extraction, pile head trimming, pile testing, and demobilization in accordance with approved drawings, project specifications, and applicable RTA, DEWA, Dubai Municipality, and HSE requirements

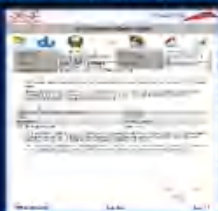
3. SUMMARY OF PLANT & EQUIPMENT, MATERIALS, TOOLS



- Plant: - Rotary Piling Rig, Crawler Crane, Vibro Hammer (if casing), Excavator, Concrete Pump, Transit Mixers, Generator, Air Compressor, Dewatering Pump.
- Tools: Survey Equipment,

- Cutting / Bending Machine, Grinder / Welder, Lifting Gear, Torque Wrench, Measuring Instruments.
- Materials: Reinforcement Cage, Temporary Casing, Tremie Pipes, Ready-Mix Concrete, Bentonite (if required), Spacers.

4. PERMITS, NOCS



- Internal Permits:
- Excavation, Lifting Permit, Hot Work (if applicable), Electrical PTW, Work at Height (if required),

- Authority NOCs:
- RTA ROW Permit, DEWA Utility NOC, DU/Etisalat NOC (if applicable)

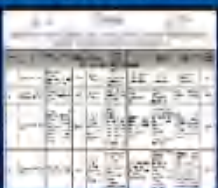
5. SUMMARY OF SPECIFIC TRAINING (INTERNAL & EXTERNAL)



- Internal Training:
- HSE Induction, MSRA / PTW Awareness,

- External/Certified:
- Crane Operator, Rigger/Banksman, Lifting Supervisor, First Aid & Fire Fighting, Heavy Equipment Operator.

6. LEGAL REFERENCE / REQUIREMENT



- UAE Federal Laws (Labour & Environment),
- DM Local Order 61/1991 (CoCSP),

- RTA QHSE MSTG
- RTA ROW & WZTM Manual,

7. SUMMARY OF PERSONAL PROTECTIVE EQUIPMENTS



- Mandatory PPEs:
- Helmet, High-Visibility Vest, Safety Shoes.

- Task-Specific:
- Hearing Protection, Gloves, Goggles, Face Shield, Dust Mask, Full Body Harness (if required).

8. SUMMARY OF INSPECTIONS

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

- Daily HSE Inspection, Excavation/Bore Stability Check, Working Platform Verification, Plant Pre-Start Inspection, Crane & Lifting Gear Certification Check, Electrical DB Inspection, Pile Integrity/Load Test Inspection, Traffic Diversion Inspection



RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

9. RISK ASSESSMENT MATRIX -

A. OH&S RISK RATING CATEGORIES & ACTION LEVEL ↓			B. DURATION OF RISK EXPOSURE & AREA OF INFLUENCE ↓			C. RISK MATRIX INTERACTION (L × S) 5×8 MATRIX ↓ (HEAT MAP)							
RISK LEVEL	SCORE RANGE	REQUIRED ACTION	RISK LEVEL	DURATION	AREA	SEVERITY 5 Level, Vertical	LIKELIHOOD - 6 levels, horizontal						
L - Low Acceptable	1-4	Self-inspection with minor controls, minimal supervision	LOW L	SHORT TERM (< 1 DAY)	LIMITED (ON-SITE)			1- REMOTE	2- UNLIKELY	3- RARE	4- LIKELY	5- COMMON	6-ALMOST CERTAIN
M - Medium Moderate	5-9	Basic controls before task starts	MEDIUM M	MEDIUM TERM (< 1 MONTH)	LIMITED (WITHIN SITE BOUNDARY)		5 Catastrophic	HIGH	HIGH	VERY HIGH	VERY HIGH	VERY HIGH	VERY HIGH
H - High Unacceptable	10-15	Before Task-Additional Controls, Commences + Management Approvals Required	HIGH H	LONG TERM (1-6 months)	EXTENSIVE (OFF-SITE Community)		4 Major	MEDIUM	MEDIUM	HIGH	HIGH	VERY HIGH	VERY HIGH
VE - Very High Prohibited	16-20	Immediate Stop Work	VERY HIGH VH	VERY LONG TERM (> 6 months)	EXTENSIVE (OFF-SITE COMMUNITY)		3 Moderate	MEDIUM	MEDIUM	HIGH	HIGH	HIGH	HIGH
							2 Minor	LOW	LOW	MEDIUM	MEDIUM	HIGH	HIGH
						1 Incidental	LOW	LOW	LOW	MEDIUM	MEDIUM	MEDIUM	
D. SEVERITY SCALE (1 - 5) ↓			E. LIKELIHOOD SCALE (1 - 6) ↓										
SEVERITY ↓	MEANING	ENVIRONMENT IMPACT	SAFETY IMPACT	HEALTH IMPACT	LIKELIHOOD ↓	MEANING	PRACTICAL DESCRIPTION	CONSTRUCTION CONTEXT					
5 CATASTROPHIC	Multiple or single fatality / major system failure	Major environmental damage, regulatory shutdown, widespread impact	Fatality or multiple fatalities	Life-threatening irreversible injury	1 REMOTE	Extremely unlikely	No known occurrence in similar controlled works	Fully engineered excavation with certified shoring					
4 MAJOR	Permanent disability or severe injury	Significant environmental impact requiring authority intervention	Permanent disability amputation	Serious medical condition requiring hospitalization	2 UNLIKELY	Possible but not expected	It has occurred in industry but rare under controls	Utility strike despite trial pits					
3 MODERATE	Medical treatment case / Lost Time Injury	Noticeable environmental impact requiring corrective action	Minor injury requiring first aid	Temporary illness or recoverable injury	3 RARE	Occasional occurrence	It has occurred in similar projects	Minor trench instability					

HAZARD IDENTIFICATION & RISK ASSESSMENT

2 MINOR	First aid case	Localized minor environmental impact	Minor injury requiring first aid	Minor reversible health effect	4 LIKELY	Expected to occur sometimes	Occurs periodically if controls not strict	Expected to occur sometimes
1 INCIDENTAL	No injury / negligible consequence	Negligible environmental impact	No injury	No health effect	5 COMMON	Occurs frequently	Repeated exposure to routine tasks	Dust exposure without suppression
					6 ALMOST CERTAIN	Will occur without control	Happens regularly in absence of controls	Slip / trip without housekeeping

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (H&S&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
1.	Site Mobilization	Illegal workers deployed. Unaware of site conditions. Unfit personnel. Lack of PPE. Lone working. Psychosocial stress. Defective equipment mobilized. Adverse weather conditions.	Personal injury. Property damage. Minor to major injury. Legal penalties (fines). Impact from extreme weather (heat, wind, rain, lightning, dust storm).	3 Likely	4 Major	12 High	Lone work is strictly prohibited. New workers paired with experienced personnel under buddy system during mobilization. Pre-employment medical fitness screening conducted. Welfare facilities including drinking water, rest shelters and first aid arrangements established prior to mobilization. Compliance with DM Summer Working Hours Regulation. Hydration stations and shaded rest areas provided. Work areas inspected after rainfall to remove water accumulation. All plants and equipment verified for valid third-party certification prior to entry. HSE department conducts pre-entry inspection and document verification. Equipment is not permitted to operate without inspection stickers and clearance. Mandatory Project HSE Induction prior to site entry covering site hazards, emergency procedures and reporting protocols. Verification of Emirates ID, visa status and workmen compensation insurance.	Conduct periodic welfare inspections and health monitoring. Re-inspect scaffolds, excavations and temporary structures after adverse weather before resuming work. Suspend work during severe weather conditions including lightning, sandstorm or high winds. Increase random document verification inspections and conduct daily pre-start briefings.	1 Remote	4 Major	4 Major	HSEO / Admin / Supervisor / Male Nurse

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							Weather conditions monitored daily and adequate lighting provided during low visibility. Mandatory PPE includes safety helmet, safety shoes and high-visibility vest worn at all times. Task-specific PPE provided as per activity requirement. Compliance enforced through supervision and induction records maintained.					

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS										DATE: 18-03-2026		
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (H&AE)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
2.	Traffic Management Setup	Vehicle intrusion into work zone Collision between vehicles / with workers / pedestrian's Pedestrian strike Vehicle overturning / loss of control Traffic congestion leading to secondary incidents	Fatality / Multiple casualties Serious injury Traffic disruption Property damage Regulatory penalties / fines	4 Likely	5 Catastrophic	20 Very High	No setup without approved RTA TMP and ROW permit. No work in live lanes without full closure/diversion. Approved TMP implemented with speed reduction (signage, rumble strips). Lane closure/diversion as per drawing. Seatbelts are mandatory for all vehicles. Concrete/water-filled barriers + TMA deployed. Pedestrian walkways are fully segregated and marked. Buffer zones maintained. Certified TTM Supervisor + trained flagmen control flow. Daily inspection of barriers/signage/TMA. Emergency response plan for traffic incidents. Pre-shift toolbox talks on hazards. Mandatory high-visibility jacket, safety helmet, safety boots. Hand gloves. All traffic personnel were inducted on TMP. Strict supervision by TTM Marshal.	Additional warning signs and flashing lights installed during night work. Deployment of shadow vehicle or TMA where required. Continuous monitoring of traffic diversion by Traffic Management Team. Immediate rectification of any displaced barriers or cones.	1 Remote	5 Catastrophic	5 Medium	Traffic Team Supervisor / TTM Marshal / HSE
3.	Survey & Setting Out	Slips, trips and falls on uneven ground; struck by moving plant; working near heavy equipment; working near live traffic; improper survey	Minor to major injury; property damage; traffic accident	3 Rare	4 Major	12 High	Survey activities shall only commence after confirmation that the working platform and access routes are safe and clear of obstruction. Survey work shall be planned and coordinated with plant	Survey work to be coordinated with piling supervisor before equipment operation begins. Temporary cones or barricades placed around survey equipment to improve visibility. Additional	1 Remote	4 Likely	4 Low	Chief Surveyor / Surveyor / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) – Section C.8

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		equipment setup; heat stress exposure					operators to avoid working within active equipment operating zones. Survey areas shall be inspected for uneven ground, loose debris, and trip hazards before starting work. Survey markers and equipment shall be positioned in stable locations away from equipment movement routes. Exposure to heat shall be minimized through proper work scheduling and rest breaks. Survey personnel shall remain within designated work zones and away from heavy equipment operating areas. Where work is close to plant movement routes, barriers or cones shall be installed to segregate surveyors from equipment. Communication shall be maintained between surveyors, supervisors, and plant operators before entering the work area. Survey equipment such as total stations shall be properly secured on stable ground. Dedicated Flagman / banksmen shall control plant movement near survey activities. Mandatory PPE including safety helmet, high-visibility vest, safety shoes, and gloves shall be worn by survey personnel. Toolbox talk shall be conducted prior to survey work highlighting hazards related to plant movement	supervision when survey work is carried out near heavy equipment or live traffic areas.				

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							and uneven ground. Survey team shall strictly follow site safety rules and instructions from supervisors and HSE personnel.					
4.	Utility Identification & Trial Trenching	Contact with underground utilities; damage to electrical cables; gas leak from pipelines; flooding due to water line rupture; excavation collapse; struck by excavation equipment; slips, trips and falls	Fatality; serious injury; service disruption; property damage; authority penalty	4 Likely	5 Catastrophic	20 Very High	Excavation and trial trenching shall only commence after obtaining approved utility drawings and relevant NOCs from concerned authorities. All underground services shall be identified through utility detection survey and marked clearly on site before excavation. Trial trenching shall be carried out manually in areas where underground utilities are expected. Mechanical excavation shall not be permitted within the safe clearance distance from known utilities. Excavation depth shall be controlled and monitored continuously to prevent accidental damage to services. Work areas shall be barricaded with warning signage to prevent unauthorized entry. Exposed utilities shall be properly protected and supported to prevent damage during excavation. Safe access and egress shall be provided around the excavation area. Permit to Dig and excavation permits shall be obtained before starting work. Trial trenching shall be carried out under supervision of the Site Engineer and HSE	Use of cable locator / ground penetrating radar where required. Additional supervision during excavation near critical utilities such as high voltage cables. Immediate stops work and notify authorities if unidentified utility is encountered.	1 Remote	5 Catastrophic	5 Medium	Construction Manager / Site Engineer / Supervisor HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
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							personnel. Dedicated flagman / banksmen shall guide excavation equipment when mechanical excavation is required. Continuous monitoring shall be maintained to identify any exposed utilities during excavation. Emergency procedures should be in place in case of utility damage or service interruption. Mandatory PPE including safety helmet, high-visibility vest, safety shoes, gloves, and eye protection shall be worn by all workers involved in excavation activities. Toolbox talk shall be conducted prior to excavation activities to brief workers about underground utility hazards and safe excavation practices. Strict compliance with Permit to Dig procedures shall be enforced by supervisors and HSE personnel.					
5.	Mobilization of Piling Plant & Equipment	Plant overturning; collision with personnel; struck-by moving equipment; blind spot hazard; unstable ground condition; mechanical failure; unauthorized access to work zone	Fatality; major injury; property damage; traffic accident	4 Likely	5 Catastrophic	20 Very High	Mobilization shall commence only after confirmation that the access route and piling working platform are properly prepared, levelled, and approved by the Project Engineer and HSE Department. Unnecessary simultaneous movement of multiple heavy equipment shall be avoided to eliminate congestion. Unauthorized personnel shall be strictly restricted from entering the plant operating zone. Ground bearing capacity of the working platform shall be verified	Engineer verification of working platform prior to positioning piling rig. Additional banksman assigned during tight access movement. Night operations are only permitted with adequate lighting and supervision.	1 Remote	5 Catastrophic	5 Medium	Construction Manager / Piling Supervisor / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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				L	S	IRR			L	S	RRR	
							and compacted to support the load of the piling rig and crawler crane. Access routes shall be inspected to ensure they are free from obstacles, soft ground, or underground utility exposure. Plant movement speed inside the site shall not exceed 20 km/h. Mobilization shall preferably be scheduled during low traffic periods. Mobilization and unloading areas shall be barricaded using rigid barriers, cones and warning signage. Pedestrian walkways shall be segregated from equipment movement routes. Dedicated banksman / Flagman shall control plant movement. Adequate lighting shall be provided during night or low visibility conditions. Only third-party certified piling rigs, cranes, and lifting equipment with valid certification shall be allowed on site. Plants shall undergo pre-mobilization inspection and daily pre-start checks. All equipment shall be fitted with reverse alarms, flashing beacons and mirrors. Only trained and authorized operators shall operate the machinery. Communication between operators and Flagman / banksman shall be maintained through standard hand signals or radio communication.					

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

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				L	S	IRR			L	S	RRR	
							Mobilization shall be carried out under supervision of the Construction Engineer and HSE personnel Mandatory PPE including safety helmet, high-visibility vest, safety shoes, gloves and eye protection shall be worn by all personnel. Toolbox talk shall be conducted prior to mobilization explaining hazards and safe movement procedures. All workers shall comply with site traffic rules and instructions of the banksman. Unsafe conditions or near misses shall be reported immediately and corrective action taken.					
6.	Working Platform Preparation	Unstable ground condition; collapse of working platform; plant overturning; contact with underground utilities; slips, trips and falls; struck by moving equipment; dust generation; unauthorized access to work area	Fatality; major injury; plant damage; utility damage; property damage	4 Likely	5 Catastrophic	20 Very High	Site establishment and working platform preparation shall only commence after approval of the Method Statement, Risk Assessment, and utility clearance. Existing underground utilities shall be identified through approved drawings, trial trenching, and utility detection surveys prior to ground preparation works. The working platform shall be properly levelled, compacted, and constructed with suitable material (approved sub-base or granular fill) to achieve sufficient bearing capacity to support piling rig loads.	Independent verification of working platforms bearing capacity prior to rig mobilization. Installation of warning signage and lighting around the working area. Continuous monitoring of ground condition especially after rainfall. Immediate repair and re-compaction of any damaged or soft platform areas.	1 Remote	5 Catastrophic	5 Medium	Construction Manager / Site Engineer / Piling Supervisor / HSE

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							Ground conditions shall be assessed by the Project Engineer before positioning heavy equipment. Dust generation during ground preparation shall be minimized by periodic water spraying. Work area shall be barricaded using rigid barriers, cones, and warning signage to prevent unauthorized entry. Access and egress routes shall be clearly defined and segregated from heavy equipment movement areas. Safe pedestrian walkways shall be established within the site. All plants and equipment involved in site preparation shall be operated by trained and authorized operators. Excavators, loaders, and rollers shall be equipped with functioning reverse alarms and beacon lights. Banksman / Flagman shall guide equipment movement to avoid collision with workers or other equipment. The working platform should be inspected by HSE team prior to mobilization of piling equipment. Daily inspection of the working platform shall be conducted to identify any settlement, soft ground, or surface damage. All personnel should wear mandatory PPE including safety					

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

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							helmet, high-visibility vest, safety shoes, gloves, and dust mask where required. Toolbox talk shall be conducted prior to site preparation activities to brief workers on hazards, safe work procedures, and emergency response measures. Strict supervision shall be maintained by the Supervisor and HSE personnel.					
7.	Rig Assembly	Rig collapse during assembly; struck-by falling parts; pinch points / crushing; working at height; heavy lifting / rigging failure; unauthorized access; electrical hazards from power tools	Fatality; major injury; permanent disability; major property damage; service disruption	4 Likely	5 Catastrophic	20 Very High	Rig assembly shall only commence after approval of the Method Statement, Risk Assessment, and verification of stable working platform. No assembly without approved lifting plan and PTW. All lifting points and rigging gear inspected and certified prior to use. Assembly sequence followed strictly as per manufacturer manual and approved lifting plan. Ground bearing capacity verified before positioning rig components. Full exclusion zone established around assembly area with rigid barriers, cones, and warning signage. Only authorized personnel are allowed inside the zone. Assembly carried out under supervision of competent Piling Supervisor and Lifting Supervisor. Certified crane operators and riggers assigned. Pre-assembly toolbox talks	Ground bearing pressure calculation and additional mats if soft/uneven ground. Stop assembly of rig if wind speed exceeds safe limit. Additional spotters/banksmen for complex lifts. Physical barriers reinforced around outriggers. Full-time Lifting Supervisor presence during assembly. Daily rig component inspection records submitted to HSE. Reinforced through pre-assembly PPE checks. Pre-assembly briefing and sign-in for all personnel involved. Immediate stop-	1 Remote	5 Catastrophic	5 Medium	Piling Supervisor / Lifting Supervisor / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS									DATE: 18-03-2026			
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							and lift briefing conducted. Communication is maintained via radio or hand signals. Emergency rescue plan in place. Mandatory safety helmet, high-visibility vest, safety shoes, gloves, full-body harness (for height work), and eye protection. All personnel inducted on rig assembly hazards. Strict supervision by Piling Supervisor and HSE personnel. Stop-work authority enforced for any unsafe condition.	work authority for any unsafe condition.				
8.	Rig Movement	Rig overturning on uneven/soft ground; collision with personnel or other plant; struck-by during movement; blind spot hazards; contact with underground utilities; mechanical failure; unauthorized access to movement route	Fatality; major injury; permanent disability; major property damage; utility damage; service disruption	4 Likely	5 Catastrophic	20 Very High	Rig movement shall only commence after confirmation that the access route and working platform are properly prepared, levelled, compacted, and approved by the Project Engineer and HSE Department. No movement without dedicated banksman / flagman. Ground bearing capacity verified and route inspected for obstacles, soft ground, or underground utilities. Plant movement speed is strictly limited to 20 km/h inside site. Movement preferably scheduled during low traffic periods. Full exclusion zone established around the rig movement path using rigid barriers, cones, and warning signage. Pedestrian walkways segregated from equipment routes.	Engineer verification of route and platform prior to movement. Additional Flagman / banksman assigned during tight or blind-spot movements. Night movement only permitted with adequate lighting and extra supervision. Installation of proximity alarms on rig where available.	1 Remote	5 Catastrophic	5 Medium	Construction Manager / Piling Supervisor / Logistics / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

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				L	S	IRR			L	S	RRR	
							Safe buffer zones maintained between rig and live traffic. Only third-party certified rig and operator allowed. Pre-movement inspection and daily pre-start checks conducted. Rig fitted with reverse alarms, flashing beacons, and mirrors. Communication is maintained via hand signals or radio between operator and banksman. Movement supervised by Piling Supervisor and HSE personnel. Mandatory safety helmet, high-visibility vest, safety shoes, gloves, and eye protection for all personnel in the vicinity. Toolbox talk conducted prior to movement explaining hazards and safe procedures. Strict supervision by Construction Manager / Piling Supervisor and HSE. Stop-work authority enforced for any unsafe condition.					
9.	Drilling Operation	Borehole instability / collapse; utility strikes; drilling fluid spills (env impact); dust / noise; struck-by rotating parts; fall into borehole; mechanical failure	Fatality; serious injury; burial; service disruption; environmental contamination; regulatory fines	4 Likely	5 Catastrophic	20 Very High	Rotary drilling shall commence only after confirmation of approved method statement, utility clearance, and permit to dig. Underground services shall be verified through utility drawings and trial trenching prior to drilling. Drilling operations shall be carried out using properly maintained piling rigs operated by trained and authorized operators.	Installation of rigid barricades or steel covers around boreholes when drilling is paused. Additional monitoring of borehole stability during drilling. Immediate suspension of work during severe weather or unstable ground conditions.	1 Remote	5 Catastrophic	5 Medium	Piling Supervisor / Site Engineer / Operator / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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							The drilling area shall be inspected to ensure the working platform is stable and capable of supporting the rig load. The drilling speed and operational parameters shall be controlled to prevent instability of the borehole. Noise and vibration exposure shall be minimized through proper equipment maintenance and work planning. The drilling area shall be barricaded to establish a clear exclusion zone around the piling rig. Unauthorized personnel shall not be allowed within the operating radius of the rig. Warning signs and safety barriers shall be installed around the open borehole to prevent accidental entry or falls. Drilling operations shall be supervised by a competent piling supervisor. Operators shall maintain constant communication with the banksman during drilling activities. The borehole condition shall be continuously monitored to detect any signs of collapse or instability. Regular inspection of drilling tools and equipment shall be conducted. Stop work procedures shall be implemented immediately if unsafe conditions are identified. Mandatory PPE including safety helmet, high-visibility vest, safety					

HAZARD IDENTIFICATION & RISK ASSESSMENT

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							shoes, gloves, safety goggles, and hearing protection shall be worn by all personnel working near drilling operations. Toolbox talk shall be conducted before starting drilling activities highlighting hazards associated with rotating machinery and open boreholes. Strict supervision shall be maintained by the Supervisor and HSE personnel to ensure safe work practices are followed.					
10.	Temporary Casing Installation	Casing collapse / misalignment; struck-by vibro-hammer / crane; noise / vibration; lifting failure; pinch points; unstable ground during installation	Fatality; crush injury; hearing loss; structural damage; service disruption	4 Likely	5 Catastrophic	20 Very High	Installation of temporary casing shall only be carried out after confirmation that the working platform is properly prepared, compacted, and approved by the Project Engineer. All lifting activities shall follow an approved lifting plan and shall be executed only by trained and certified personnel. The piling rig and lifting equipment shall be inspected prior to operation to ensure they are in good working condition. The weight and dimensions of casing sections shall be verified to ensure compatibility with lifting equipment capacity. Equipment movement and lifting activities shall be planned to avoid congestion and unnecessary lifting hazards. A clearly defined exclusion zone shall be established around the	Casing Installation activities to be stopped immediately during high wind conditions or unstable ground conditions. Additional monitoring of rig stability during casing installation. Regular inspection of lifting equipment and rig components during operation.	1 Remote	5 Catastrophic	5 Medium	Piling Supervisor / Site Engineer / Lifting Supervisor / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							piling rig during casing installation to prevent unauthorized personnel from entering the operating area. Barricades, cones, and warning signage shall be installed to isolate the lifting zone. Only authorized personnel involved in the operation shall be allowed within the work area. Installation of casing shall be carried out under supervision of a competent piling supervisor. Certified crane operators and riggers shall be assigned for lifting activities. Lifting gears such as slings and shackles shall be inspected and certified before use. The vertical alignment of casing shall be monitored to ensure stability during installation. All personnel involved wear mandatory PPE including safety helmet, high-visibility vest, safety shoes, gloves, and eye protection. Hearing protection shall be used where required due to noise from drilling operations. Toolbox talk shall be conducted prior to casing installation to brief workers about hazards related to suspended loads, rotating equipment, and rig operations. Strict supervision shall be maintained by the Supervisor and HSE personnel to ensure compliance with safe working procedures.					

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

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				L	S	IRR			L	S	RRR	
11.	Rebar Cage Fabrication	Manual handling strains; sharp edges / cuts; falling rebar / dropped bars; struck-by during lifting / movement; welding / hot work sparks; poor housekeeping; repetitive strain.	Musculoskeletal injury; lacerations / amputation; crush injury; fire / burns; respiratory irritation	4 Likely	4 Major	16 Very High	Reinforcement fabrication shall be carried out only in designated fabrication areas away from heavy plant movement and piling operations. Work areas shall be properly organized to eliminate congestion and unnecessary material handling. Reinforcement bars shall be stored properly on racks or timber supports to prevent rolling or falling. Mechanical cutting and bending equipment shall be used to minimize manual handling. Work surfaces shall be kept clean and free of scrap materials to reduce trip hazards. The fabrication area should be clearly barricaded or marked to restrict access by unauthorized personnel. Welding and cutting operations shall be carried out within designated areas away from other work activities. Rebar cutting and bending machines should be inspected before use to ensure they are in safe working conditions. Guards and safety devices on equipment shall not be removed. Only trained and authorized workers shall operate fabrication equipment. Adequate fire extinguishers shall be available in areas where cutting or welding activities are performed.	Installation of rebar caps on exposed reinforcement ends. Regular inspection of cutting and bending equipment. Additional supervision during welding operations to prevent fire hazards.	1 Remote	4 Major	4 Low	Foreman / Fabrication Supervisor / HSE

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
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				L	S	IRR			L	S	RRR	
							All workers shall wear mandatory PPE including safety helmet, high-visibility vest, safety shoes, gloves, and safety goggles. Face shields should be used during cutting or grinding operations. Toolbox talk shall be conducted prior to fabrication activities to inform workers about hazards related to sharp edges, sparks, and equipment operation. Workers shall follow proper housekeeping practices and maintain clear walkways within the fabrication area.					
12.	Rebar Cage Lifting	Load swing / uncontrolled movement; rigging failure / sling snap; load drop / crush; crane / HIAB overturning; swinging load striking personnel or structures; boom contact with overhead lines; ground failure / outrigger settlement	Fatality; crush injury; permanent disability; major property damage; service disruption	4 Likely	5 Catastrophic	20 Very High	No lifting over personnel or occupied areas. No lifting without approved lifting plan and permit. Approved lifting plan specific to cage weight, crane, and site conditions. Certified crane and lifting gear (third-party verified, colour-coded, inspected). Outriggers fully extended on firm ground with mats/pads. Load weight verified within crane capacity. Taglines used to control swings. Full exclusion zone established, barricaded, and signed around lift area. Rigger / signaler with clear line of sight. Overhead power line clearance checked and prohibited zones marked. Lifting permit / PTW required. Competent, certified operator, rigger,	Trial lift performed for critical/long cages. Additional spotters for blind or complex lifts. Ground bearing pressure check and extra mats if soft ground. Stop lifting if wind exceeds manufacturer limit. Full-time Lifting Supervisor presence. Continuous radio communication. Daily crane inspection log and photo records. PPEs reinforced through pre-lift checks. Immediate stop-work authority for any unsafe condition.	1 Remote	5 Catastrophic	5 Medium	Lifting Supervisor / Rigger / Banksman / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS										DATE: 18-03-2026		
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							and flagman deployed. Pre-lift briefing and trial lift for critical loads. Emergency lowering / rescue plan in place. Mandatory safety helmet, full body harness (for riggers/flagman at height), high-visibility vest, safety boots, gloves. All lifting personnel inducted on hazards and hand signals. Strict supervision by Lifting Supervisor. No unauthorized entry into exclusion zone.					
13.	Lowering Cage	Load drop / uncontrolled swing; struck-by falling cage; pinch points during lowering; collision with borehole wall; fall into open borehole; rigging failure; loss of control during descent	Fatality; crush injury; permanent disability; major property damage; borehole collapse	4 Likely	5 Catastrophic	20 Very High	No lowering without approved lifting plan and PTW. No personnel allowed under suspended cage or inside borehole during lowering. Cage weight, lifting points, and crane capacity verified before lowering. Taglines used to control swing and rotation. Load lowered slowly and vertically with continuous banksman guidance. Full exclusion zone established around borehole and lifting area with rigid barriers and signage. Only authorized personnel (rigger, banksman, operator) are allowed inside zone. Control: Lifting permit / PTW required. Competent certified operator, rigger, and banksman deployed. Pre-lowering briefing and trial lift performed. Continuous radio	Trial lowering (dry run) for critical cages. Additional spotters for blind or complex lowering. Ground bearing pressure check and extra mats if soft ground. Stop lowering if wind exceeds manufacturer limit. Full-time Lifting Supervisor presence. Continuous radio communication. Daily crane inspection log and photo records. PPEs reinforced through pre-lowering checks. Immediate stop-work authority for any unsafe condition.	1 Remote	5 Catastrophic	5 Medium	Lifting Supervisor / Rigger / Banksman / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
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							communication is maintained. Emergency lowering procedure in place. Mandatory safety helmet, full body harness (for riggers at height), high-visibility vest, safety boots, gloves. All personnel inducted on lowering hazards and hand signals. Strict supervision by Lifting Supervisor. Stop-work authority enforced for any unsafe condition.					
14.	Tremie Pipe Installation	Pipe drop / uncontrolled swing during lifting. Struck-by falling pipe sections. Pinch points / crushing during joining; Concrete splash / burns when pumping starts. Fall into open borehole. Manual handling strains; rigging failure	Fatality. Crush injury; chemical burns; permanent disability; major property damage; environmental spill	4 Likely	5 Catastrophic	20 Very High	No tremie pipe installation without approved lifting permit and it must be in the stand frame in vertical position. No personnel allowed under suspended pipe or inside boreholes during assembly and lowering. Pipe sections weight and lifting points verified before lifting. Certified slings and shackles used. Joints inspected and cleaned prior to connection. Load lowered slowly and vertically with continuous banksman guidance. Full exclusion zone established around borehole and lifting area with rigid barriers and signage. Only authorized personnel (rigger, banksman, operator) are allowed inside zone. Lifting permit / PTW required. Competent certified operator, rigger,	Trial lowering (dry run) for critical pipe lengths. Additional spotters for blind or complex lowering. Ground bearing pressure check and extra mats if soft ground. Stop lowering if wind exceeds manufacturer limit. Full-time Pour Supervisor presence. Continuous radio communication. Daily crane/pipe inspection log and photo records. PPEs reinforced through pre-assembly checks. Immediate stop-work authority for any unsafe condition.	1 Remote	5 Catastrophic	5 Medium	Tremie Pipe Installation

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

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				L	S	IRR			L	S	RRR	
							and banksman deployed. Pre-assembly briefing and trial lift performed. Continuous radio communication is maintained. Emergency lowering procedure in place. Mandatory safety helmet, full body harness (for riggers at height), high-visibility vest, safety boots, chemical-resistant gloves, face shield/goggles. All personnel inducted on tremie hazards and hand signals. Strict supervision by Pour Supervisor and Lifting Supervisor. Stop-work authority enforced for any unsafe condition.					
15.	Concrete Pumping	Concrete splash / chemical burns. Hose burst / whip effect; pump failure / over-pressurization. Struck-by flying debris. Fall into borehole. Environmental spill. Manual handling strains Formwork collapse	Fatality; serious chemical burns; crush injury; permanent disability; environmental contamination; regulatory fines	4 Likely	5 Catastrophic	20 Very High	No concrete pour without inspection of pump, pipeline, clamps, and tremie joints. No standing directly under tremie/hose joints. All clamps secured; pressure rating verified; tremie joints checked before pour. Communication protocol between pump operator and pour supervisor established. Pour area demarcated, only essential crew within controlled zone. Pump pressure monitored continuously; controlled start / stop. Emergency shut off identified and briefed. Follow supervisor's instructions on pumping speed.	Additional hose restraint / whip checks where needed. Pre-pour checklist and hold point sign-off by Site Engineer / HSE. Immediate eyewash / washing station available at pour area.	1 Remote	5 Catastrophic	5 Medium	Pour Supervisor / Pump Operator / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

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							Assign a spotter to monitor overflow risk. Safe access is maintained; housekeeping enforced to prevent slips. Manual handling minimizes using mechanical aids. Face shield/goggles, chemical-resistant gloves, long sleeves / coveralls, safety boots. Supervision maintained; stop work if leakage, unusual vibration, or clamp movement observed. Check formwork design before pouring, proper shoring/bracing, supervise concrete placement, training for workers, Never stand directly under freshly poured formwork. Report any cracks, deflection, or unusual movement immediately.					
16.	Casing Removal	Concrete lift / contamination; causing damage / collapse; worker entrapment / pinch points; rig instability; struck-by during extraction; loss of concrete head	Fatality; serious injury; reduced pile section; environmental spill; regulatory fines	4 Likely	5 Catastrophic	20 Very High	Casing removal shall only commence after completion of concrete and confirmation that concrete remains workable. No personnel allowed near casing during extraction. Concrete level maintained inside casing during extraction to exceed external pressure. Extraction performed gradually and axially. Casing joints checked for watertightness before operation.	Additional supervision during extraction stage (highest risk). Hold-point checklist before extraction (bore stability confirmed). Crew positioned, exclusion zone intact). Immediate stoppage if anyone enters pinch zone.	1 Remote	5 Catastrophic	5 Medium	Piling Supervisor / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

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				L	S	IRR			L	S	RRR	
							Full exclusion zone established around the rig and casing with rigid barriers and signage. Only authorized personnel (operator, banksman) allowed inside zone. Extraction carried out under supervision of competent piling supervisor. Rig stability ensured (mats/outriggers/level). Continuous monitoring of bore condition and concrete level. Communication is maintained via radio or hand signals. Emergency response plan in place. Mandatory safety helmet, high-visibility vest, safety shoes, gloves, and eye protection. Hearing protection where required due to rig noise. Toolbox talk conducted prior to extraction briefing workers on hazards. Strict supervision by Piling Supervisor and HSE personnel. Stop-work authority enforced for any instability.					
17.	Abortive Boreholes	Open borehole collapse. Fall into unfilled holes; grout/concrete spill during backfilling; utility damage during filling; manual handling strains; environmental	Fatality; serious injury; burial; environmental pollution; regulatory fines	4 Likely	5 Catastrophic	20 Very High	Abortive borehole shall only be treated after confirmation by the piling supervisor and PTW approval. No entry into open borehole at any time. Boreholes filled with approved grout or lean concrete mix in controlled manner. Fill material quantity and quality verified before pouring.	Installation of steel cover plates or rigid covers immediately after decision to abandon borehole. Additional supervision and photo records during backfilling. Immediate cleanup of any grout spill.	1 Remote	5 Catastrophic	5 Medium	Piling Supervisor / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		contamination; struck-by falling fill material					Manual handling minimized using mechanical aids (pumps/hoppers). Full exclusion zone established around open boreholes with rigid barriers, cones, warning signage, and steel covers where required. Only authorized personnel are allowed inside zones. Filling carried out under supervision of competent piling supervisor. Continuous monitoring of borehole stability during backfilling. PTW and pre-task toolbox talk conducted. Emergency response plan for collapse or spill in place. Mandatory safety helmet, high-visibility vest, safety shoes, gloves, dust mask/respirator, and chemical-resistant boots where grout is used. All personnel inducted on abortive borehole hazards. Strict supervision by Piling Supervisor and HSE personnel. Stop-work authority enforced for any instability.					
18.	Slurry Handling	Slurry Spill / Leak During Handling. Slips / Falls on Wet Surfaces. Chemical Exposure / Skin Burns.	Environmental Contamination. Chemical Burns. Slips / Falls. Respiratory Irritation.	4 Likely	5 Catastrophic	20 Very High	Slurry handling shall only be kept within designated bunded area. No uncontrolled discharge or spillage allowed. MSDS available and followed. Spill prevention trays and proper hose connections are used. Workers use mechanical aids	Scheduled inspection of hoses / pumps. Waste tracking records maintained. Environmental monitoring and reporting of any spills.	1 Remote	5 Catastrophic	5 Medium	Site Engineer / Supervisor / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		Environmental Contamination (Soil / Water). Manual Handling Strains. Inhalation Of Fumes	Regulatory Fines				(pumps, hoses) to minimize manual handling. Bunded / slurry pit area fully barricaded with warning signage. Restricted access to authorized personnel only. Spill kits and absorbent materials immediately became available. Housekeeping and immediate cleanup enforced. Controlled discharge and authorized disposal only. Daily inspection of hoses, pumps and bunding. Mandatory chemical-resistant gloves, goggles / face shield, boots, coveralls. All personnel inducted on slurry hazards and spill response. Strict supervision by Supervisor. Stop-work authority enforced for uncontrolled spillage.					
19.	Dewatering	Pump failure; short circuit; fire; pipeline rupture; oil spill; overflow; sediment discharge; noise; flooding of excavation	Fatality; serious injury; drowning; property damage; environmental contamination; authority fines; public disturbance	4 Likely	5 Catastrophic	20 Very High	Valid discharge permits and approved layout obtained prior to operation. Work stopped immediately if excavation floods. Backup pumps and generators are always available. Continuous monitoring of discharge flow, pressure and water level with auto cut-off. Rock fill placed at discharge point to prevent erosion and sediment discharge.	Spill kits and absorbent materials kept at discharge and fuel points. Emergency response plan activated immediately on any abnormality. 24/7 monitoring log reviewed daily by Construction Manager. Additional dewatering capacity / backup pumps during heavy rainfall.	1 Remote	5 Catastrophic	5 Medium	Pump Operator / Supervisor / Electrician / Construction Manager / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							Fuel storage in designated bunded and ventilated area with spill kits. Secondary sump / holding pit provided for excess seepage. All pumps, generators and electrical panels third-party inspected and certified. Electrical connections are done only by competent electricians. ELCB protection and proper earthing verified. Pre-commissioning QA/QC inspection carried out. Fire extinguishers were placed near generator and DB. Noise barrier installable where required. 24/7 visual inspection log maintained. Emergency shutdown system in place. Hearing protection, safety boots, gloves and high-visibility vest mandatory for operators. Only competent and trained pump operator team allowed. Pre-task briefing conducted and recorded prior to every shift. Strict supervision by Supervisor / Electrician.					
20.	Using Man Basket / Man Cage	Man-basket structural failure; overloading; fall of personnel from height; collision with nearby structures; crane or lifting equipment failure;	Fatality; serious injury; crushing injury; dropped objects; property damage; regulatory penalties	4 Likely	5 Catastrophic	20 Very High	Use of man-basket avoided wherever safer access methods such as scaffold platform, MEWP or temporary working platform are reasonably practicable. Personnel lifting is allowed only with written authorization and approved lifting plan.	Tools and materials inside basket secured with tool lanyards to prevent falling objects. Taglines are used to control basket movement and prevent swinging.	1 Remote	5 Catastrophic	5 Medium	Lifting Supervisor / Crane Operator / Banksman / Site

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		communication failure between operator and banksman; high wind causing uncontrolled basket movement; falling tools or materials; improper attachment to crane hook					Only certified man-basket designed for personnel lifting used. Basket clearly marked with Safe Working Load (SWL). All lifting accessories including crane, slings, shackles and hooks EIAC approved third-party inspected and certified. Pre-use inspection conducted prior to every lifting operation. Lifting zone barricaded with rigid barriers and warning signage. Exclusion zone maintained below suspended basket and unauthorized personnel strictly restricted from the area. Permit-to-lift and approved lifting plan implemented prior to personnel lifting. Competent lifting supervisor assigned to monitor the operation and trained banksman deployed to maintain communication with crane operator through hand signals or radio. Crane positioned on level and stable ground with outriggers fully deployed and verified. Trial lift conducted before lifting personnel. Emergency rescue plans are prepared and communicated to the team. Personnel inside the basket required to wear full body safety harnesses attached to approved anchor point within basket, safety	Weather conditions continuously monitored and lifting stopped during strong winds, heavy rain or poor visibility. Additional communication checks conducted before lifting to verify signals between crane operator and banksman. Periodic inspection of man-basket condition carried out, and maintenance records maintained.				Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							helmet with chin strap, gloves, safety footwear and high-visibility vest. Only trained and authorized personnel permitted inside the basket. Workers shall always remain inside guardrails and shall not lean outside or climb on basket structure. Strict compliance with lifting supervisor instructions maintained.					
21.	Hot Work	Fire / explosion from sparks or hot surfaces; Burns to personnel; Toxic fumes / poor ventilation; Gas leak from cylinders; Fire spread to adjacent materials; Unauthorized hot work	Fatality / Multiple burns; Major fire / property damage; Respiratory injury / asphyxiation; Explosion / blast injury	4 Likely	5 Catastrophic	20 Very High	No hot work permitted without Hot Work Permit. No hot work in areas with flammable / explosive atmospheres. Certified welder / hot work operator only. Gas cylinders fitted with flashback arrestors & regulators. Combustible materials removed or protected (covers, fire blankets) within 10 m radius. Fire-resistant barriers / screens used to contain sparks. Dedicated fire watcher deployed throughout hot work & for 30 minutes after cessation. Hot work area barricaded & signed as exclusion zone. Gas cylinders secured upright & stored away from ignition sources. Hot Work Permit system (valid for specific location, time & task).	Use of low spark / cold cutting alternatives, where technically feasible. Additional fire-resistant blankets & water hose on standby. Extra fire watcher for high-risk or prolonged hot work. Physical separation (temporary fire-resistant partitions) for adjacent areas. 30-minute fire watch post-hot work with signed log. Gas cylinder inspection & leak test before each use. Hot Work Supervisor full-time presence during operation. Daily observation records of permit, fire watch & area clearance. PPEs reinforced through	1 Remote	5 Catastrophic	5 Medium	Hot Work Supervisor / Permit Issuer / Fire Watcher / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HSE)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							Pre-task toolbox talks on fire hazards, emergency procedures & extinguisher use. Fire extinguishers (CO₂ & dry powder) & fire blanket immediately available. Gas leak test (soap solution) on connections before ignition. Ventilation / fume extraction provided in enclosed spaces. Mandatory: welding shield / face shield, flame-resistant clothing, welding gloves, safety boots, respirator (if fumes present). All personnel inducted on hot work permit & fire safety. Strict supervision by Hot Work Supervisor / Permit Issuer.	pre-use fit checks & PPE register. Pre-hot work briefing & permit sign-off for all involved. Random HSE inspection on permit compliance & fire watch enforcement.				
22.	Power Tools	Electrical shock / electrocution; unguarded blade / disc; flying debris / kickback; tool failure / breakage; entanglement in rotating parts; noise / vibration exposure	Major injury / Amputation; Electrocution / fatality; Eye / face injury; Permanent hearing / hand-arm vibration syndrome	4 Likely	5 Catastrophic	20 Very High	No use of defective / damaged tools. All tools fitted with guards / blade covers (no removal allowed). Cable management (no trailing leads, secured & routed safely). Pre-use visual inspection & function test by user. Low-vibration tools selected where available. ELCB protection on all circuits / distribution boards. Tools colour-coded & tagged. Tool storage in locked cabinets when not in use.	Use of battery-powered tools (cordless) for high-risk tasks where feasible. Anti-vibration gloves for prolonged use. Additional ELCB Testing & tagging before each shift. Dedicated tool inspection station with lock-out for faulty items. Daily tool issue / return log with user sign-off. Supervisor full-time presence during high-risk tool use (e.g., angle grinders, saws). Vibration exposure records	1 Unlikely	5 Catastrophic	5 Medium	Supervisor / Electrical Officer / Trade Foreman / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							Portable power tools register maintained (inspection & issue log). Pre-use toolbox talk covering safe operation, PPE, and emergency stop. Noise / vibration monitoring for prolonged use. Safety goggles / face shield, cut-resistant gloves, ear plugs / muffs, dust mask (for grinding / cutting), safety helmet, high-visibility vest. Strict supervision by Supervisor / Trade Foreman. Immediate removal & quarantine of defective tools.	& health surveillance for frequent users. PPEs reinforced through pre-use PPE fit checks. Discipline - Pre-task briefing on tool-specific hazards. Random HSE spot checks on tool condition & guard integrity.				
23.	Use of Temporary Electricity	Electrical shock / electrocution; short circuit / overload; damaged cables / poor insulation; improper grounding; fire from electrical faults; unauthorized use / tampering; water contact with live cables	Fatality; serious burns; fire / explosion; major property damage; regulatory fines	4 Likely	5 Catastrophic	20 Very High	No temporary electrical installation without approved design and Electrical PTW. No live work without isolation and lock-out/tag-out. All cables routed safely, protected from damage and water. Distribution boards fitted with ELCB / RCD protection. Overload protection devices installed. Cables are properly sized and rated for load. Electrical panels and generators are placed in dedicated, secure, banded areas away from water and traffic. Live cables are clearly marked and protected with conduits or covers.	Additional ELCB testing and tagging. Dedicated electrical supervisor full-time presence during high-load periods. Regular thermographic inspection of panels. Immediate isolation and repair of any damaged cable. PPEs reinforced through daily pre-use checks.	1 Remote	5 Catastrophic	5 Medium	Electrical Engineer / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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				L	S	IRR			L	S	RRR	
							All temporary electrical work carried out by competent certified electricians. Pre-installation inspection and testing of DB, cables and earthing. Daily visual checks and weekly testing of ELCB. PTW system enforced. Emergency shut-off switches clearly marked and accessible. Mandatory safety helmet, high-visibility vest, safety shoes, insulated gloves, and face shield where required. All personnel inducted on electrical hazards and "no touch" policy. Strict supervision by Electrical Engineer / HSE. Stop-work authority enforced for any unsafe condition.					
24.	Crane Lifting	Load drop / rigging failure; crane overturning; swinging / uncontrolled load; boom contact with overhead lines / structures; ground failure / outrigger settlement; collision with personnel / plant	Fatality; crush injury; permanent disability; major property damage; service disruption	4 Likely	5 Catastrophic	20 Very High	No lifting over personnel or occupied areas. No lifting without approved lifting plan and permit. Approved lifting plan specific to load, crane, and site conditions. Certified crane and lifting gear (third-party verified, colour-coded, inspected). Outriggers are fully extended on firm, level ground with steel / timber mats. Load weight verified and within crane capacity (80% max for critical lifts). Taglines used to control load swing. Full exclusion zone established, barricaded, and signed around lift area. Wind speed monitoring	Trial lift performed for critical loads. Additional Rigger / banksmen for complex or blind lifts. Ground bearing pressure check and extra mats if soft ground. Stop lifting if wind exceeds manufacturer safe limit. Full-time Lifting Supervisor presence. Continuous radio communication.	1 Remote	5 Catastrophic	5 Medium	Lifting Supervisor / Crane Operator / Rigger / Flagman / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS										DATE: 18-03-2026		
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							(anemometer on site). Flagman / signaler with clear line of sight. Overhead power line clearance checked and no-go zones marked. Lifting permit / PTW required for all crane lifts. Competent, certified operator, rigger, and flagman deployed. Pre-lift briefing and trial lift (dry run) for critical / non-routine lifts. Emergency lowering / rescue plan in place. Safety helmet, full body harness (for riggers / flagman at height), high-visibility clothing, safety boots, gloves. All lifting personnel inducted on crane hazards and hand signals. Strict supervision by Lifting Supervisor during operation. No unauthorized personnel in exclusion zone.	Daily crane inspection log. PPEs reinforced through pre-lift checks. Immediate stop-work authority for any unsafe condition.				
25.	Manual Handling	Improper lifting / carrying posture; Overexertion / heavy loads; Repetitive strain / awkward postures; Dropping load / struck-by falling object; Slips / trips while carrying	Musculoskeletal injury / back strain; Hernia / permanent disability; Fracture / crush injury from dropped load; Acute / chronic pain	3 Rare	4 Major	12 High	No manual handling of loads >25 kg without mechanical aid. Manual handling training provided (correct posture, team lifting). Use of mechanical aids (trolleys, wheelbarrows, hoists, forklift) for heavy / bulky items. Job rotation & scheduled breaks for repetitive tasks. Loads assessed & split into smaller units where possible. Exclusion zone around heavy lifting / carrying operations.	Additional mechanical aids (pallet trucks, lifting straps) introduced for repetitive tasks. Job rotation & task redesign to minimize awkward postures. Dedicated lifting / carrying paths marked & kept clear.	1 Remote	4 Major	4 Low	Supervisor / HSE Officer / Trade Foreman

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HSE)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							Stable, level ground ensured for carrying paths. Pre-task toolbox talks on manual handling risks & safe techniques. Load weight & handling assessment before task (TILE principle - Task, Individual, Load, Environment). Immediate reporting of discomfort / pain. Health surveillance for frequent handlers. Mandatory: safety helmet, safety boots, gloves. Additional: back support belt. All personnel inducted on manual handling & "ask for help" policy. Strict supervision by Supervisor / Trade Foreman.	Supervisor daily check for correct technique & aid usage. Manual handling risk assessment updated for new materials. Health surveillance follow-up for any reported discomfort. PPEs reinforced through daily compliance checks. Discipline - Pre-shift briefing on manual handling & team lifting. Random HSE spot checks on posture & aid usage.				
26.	Refueling	Fuel spill / leak during refueling; Fire / explosion from ignition sources (sparks, hot surfaces, smoking); Static electricity discharge; Slips on spilled fuel; Inhalation of fuel vapors; Vehicle / plant movement during refueling	Fatality / serious burns (fire/explosion); Environmental contamination (soil/water pollution); Respiratory irritation / poisoning; Slips / falls / minor to major injury; Regulatory fines / shutdown	4 Likely	5 Catastrophic	20 Very High	No refueling with engine running. No refueling in areas with open flames, hot work, or smoking. No refueling during electrical storms / lightning risk. Designated refueling zones / bunded areas with spill containment. Fuel bowsters / pumps fitted with automatic shut-off nozzles & overfill prevention. Fuel stored in approved, labelled containers / tanks with secondary containment. Static grounding	Use of self-bunded mobile fuel tanks / double-walled bowsters. Automatic fuel level monitoring / alarms on storage tanks. Additional physical barriers (temporary fencing) & flashing lights around refueling point.	1 Remote	5 Catastrophic	5 Medium	Fuel Supervisor / Logistics / HSE / Plant Dept

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							straps / bonding used during transfer. Drip trays / absorbent pads placed under nozzles. Exclusion zone (cones, barriers, signage) around refueling point (minimum 3 m radius). No other plant / personnel allowed in zone during refueling. Fire extinguishers (dry powder / foam) & spill kits immediately available. Refueling Permit / checklist required (pre-check engine off, no ignition sources). Pre-refueling toolbox talks on spill/fire risks & emergency response. Daily inspection of fuel lines, nozzles, grounding & bunding. Spill response procedure (contain, absorb, report, remediate). Fuel inventory log & reconciliation. Chemical-resistant gloves, safety goggles / face shield, flame-resistant clothing (if high volume), safety boots, high-visibility vest. All refueling personnel inducted on fuel hazards & static risks. Strict supervision by Fuel Supervisor / Logistics.	Extra spill containment trays / absorbent booms. Fuel Supervisor full-time presence & sign-off during refuelling. Daily photo records of bunding, grounding & spill kit readiness. Fuel spill drill / simulation quarterly. Environmental monitoring (soil sampling) after any minor spill. PPEs reinforced through pre-refuelling PPE checks. Discipline - Pre-refuelling briefing & checklist sign-off. Random HSE inspections on grounding continuity & no-engine rule.				

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
27.	Working Near Live Traffic	Vehicle intrusion into work zone; struck-by moving vehicles; unauthorized pedestrian crossing; improper traffic diversion; collision between plant and public vehicles; poor visibility during night work	Fatality; major injury; traffic accident; property damage; regulatory penalty	4 Likely	5 Catastrophic	20 Very High	No work activities permitted in live traffic lanes without full closure / diversion. Traffic diversion and work zone setup shall only commence after approval of the Traffic Management Plan and ROW permit from RTA. Traffic flow speed near the work zone shall be reduced through advance warning signage, speed limit boards, rumble strips, and traffic calming measures. Traffic diversion layouts shall be implemented as per approved drawings and maintained throughout the work period. Work areas shall be segregated from live traffic using concrete barriers, water-filled barriers, cones, and reflective barricades. Safe pedestrian access routes shall be established and clearly marked. Adequate buffer zones should be maintained between live traffic and work activities. Certified Traffic Management Supervisor and trained flagmen shall control traffic movements at the work zone. All traffic control devices including warning signs,	Additional warning signs and flashing lights installed during night work. Deployment of TMA or shadow vehicles were required. Continuous monitoring of traffic diversion by Traffic Management Team. Immediate rectification of any displaced barriers or cones.	1 Remote	5 Catastrophic	5 Medium	Traffic Manager / Construction Manager / Supervisor / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							flashing beacons, and reflective cones shall comply with RTA Work Zone Traffic Management Guidelines. Regular inspection of traffic diversion setup shall be conducted to ensure proper alignment and visibility. Nightworks shall be supported with adequate lighting and reflective signage. All personnel working near live traffic shall wear mandatory PPE including high-visibility vest, safety helmet, safety shoes, and gloves. Toolbox talk shall be conducted for workers regarding hazards associated with working near live traffic. Strict compliance with traffic management procedures shall be enforced by supervisors and HSE personnel. Unauthorized pedestrian crossings or jaywalking within the work zone shall be strictly prohibited.					
28.	Working in Adverse Weather	Heat stress / heat stroke (high temperature / humidity) Sandstorm / dust storm (poor visibility, respiratory irritation) Lightning	Heat stroke / dehydration / Respiratory illness / eye irritation Lightning injury / electrocution Slips / falls / drowning in flooded	4 Likely	4 Major	16 Very High	No work permitted during severe weather warnings (lightning, sandstorm > certain visibility, wind > safe limit). No outdoor work during peak summer heat hours without approval.	WBGT monitoring with automated alerts & work/rest regimen adjustment. Additional shaded canopies & portable cooling units in work zones. Additional	1 Remote	4 Major	4 Low	HSE Officer / Site Supervisor / Male Nurse / Project

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		strike / electrical hazard Heavy rain / flooding / slippery surfaces High wind (falling objects, instability)	excavation Struck-by falling material / plant overturn				Daily weather monitoring via official forecasts & on-site WBGT meter. Compliance with Dubai Municipality Summer Working Hours regulation (midday break enforced). Hydration stations, shaded rest areas, and cooling facilities (fans, misting) provided. Work scheduled to avoid peak heat / adverse periods where possible. Slippery surface signage & non-slip mats in wet areas. Work suspended & personnel sheltered during lightning, sandstorm, or high wind. Emergency shelter / muster points established. Lightning protection (if tall structures involved). Heat stress awareness campaign & toolbox talks. Fatigue & heat symptom monitoring (self-reporting & buddy system). Emergency response plan for heat illness / lightning strike / flooding. Wind speed monitoring for lifting / height work (stop if exceeds limit). Mandatory: high-visibility vest, safety helmet, safety boots. Additional: heat-resistant gloves, cooling vests / neck wraps, dust mask / goggles during sandstorm. All personnel inducted on heat stress, lightning & weather hazards. Strict supervision &	drainage pumps & sump pits for heavy rain / flooding prevention. Re-inspection of scaffolds, formwork, excavations after weather event. Heat stress & lightning awareness posters / daily reminders. Emergency cooling stations (ice packs, oral rehydration) on site. PPEs - Reinforced through daily checks & stock replenishment. Discipline - Pre-shift weather & heat briefing. Random HSE inspection on hydration compliance & shelter usage.				Management

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
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				L	S	IRR			L	S	RRR	
29.	Pile Head Breaking	Flying debris / concrete fragments; Vibration exposure / hand-arm vibration syndrome; Dust inhalation; Noise exposure (>85 dBA); Struck-by falling pieces; Sharp rebar ends after breaking; Tool kickback / failure; Slips on wet concrete slurry	Eye / face injury; Respiratory illness; Permanent hearing loss; Hand-arm vibration syndrome; Lacerations / crush injury; Regulatory fines for dust / noise	4 Likely	4 Major	16 Very High	enforcement of stop-work triggers by HSE / Site Supervisor. Eliminate: Pile head breaking shall only commence after concrete has gained adequate strength (minimum 2-3 days cure or as per spec) and the area is cleared for safe access. Non-essential personnel shall be restricted from the work zone. Mechanical breaking tools (hydraulic breakers or pneumatic chipping hammers) are used to minimize manual effort. Dust generation controlled through continuous water spraying / suppression. Vibration exposure is limited by using low-vibration tools and job rotation. The pile trimming area shall be fully barricaded with rigid barriers, cones, warning signage, and debris nets / screens to contain flying fragments. Exclusion zone established around each pile being broken. Power tools inspected before use and operated only by trained and authorized workers. Adequate ventilation maintained in confined areas. Continuous supervision	Installation of protective screens or debris nets if breaking near public areas or other work fronts. Periodic rest breaks to reduce vibration exposure. Additional water suppression system / misting for dust control. Daily tool inspection log and photo records of PPE compliance.	1 Remote	4 Major	4 Medium	Supervisor / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

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RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							provided by Supervisor. Noise monitoring and zoning (>85 dBA areas marked). Pre-task toolbox talk highlighting hazards of flying debris, vibration, and dust. Mandatory safety helmet, safety goggles / face shield, cut-resistant gloves, hearing protection (ear plugs / muffs), dust mask / respirator (FFP3), safety boots. All workers inducted on pile head breaking hazards. Strict supervision by Supervisor and HSE personnel. Immediate stop-work authority enforced for any unsafe condition or PPE non-compliance.					
30.	Pile Load Testing	Overload / structural failure during test; jack / reaction system failure; sudden load release; fall from test setup; struck-by falling equipment; ground settlement / instability; electrical hazards (hydraulic pumps)	Fatality; serious crush injury; permanent disability; major property damage; test pile damage; regulatory fines	3 Rare	5 Catastrophic	15 HIGH	No load testing without approved testing procedure, reaction system design, and PTW. Non-essential personnel shall be removed from the testing area. Hydraulic jacks, gauges, and load cells inspected, calibrated, and certified before use. Load applied gradually and monitored continuously. Reaction system (kentledge / anchor piles) designed and verified by competent engineer. Ground bearing capacity checked with additional mats if soft soil. I	Additional physical barriers / fencing around test setup. Immediate stop-work if abnormal displacement or equipment failure observed. Continuous real-time data logging with alarm thresholds. Daily photo records of setup and calibration. PPEs reinforced through pre-test checks.	1 Remote	5 Catastrophic	5 Medium	Testing Engineer / Construction Manager / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							Full testing zone barricaded with rigid barriers, cones, warning signage, and exclusion tape. Adequate safety distances maintained around test setup. Only authorized testing personnel allowed inside zone. Testing supervised by qualified testing engineer. Continuous monitoring of load, displacement, and settlement throughout test. Emergency stop procedures and rescue plan established. Pre-test briefing and toolbox talk conducted covering hazards and signals. Safety helmet, high-visibility vest, safety shoes, gloves, full-body harness (for any height work on setup). All testing personnel inducted on load test hazards. Strict supervision by Testing Engineer and HSE. Stop-work authority enforced for any abnormal displacement, equipment failure, or unsafe condition.					
31.	Lighting for Night Work	Inadequate illumination leading to slips / trips / falls; struck-by moving plant /	Fatality; serious injury; crush injury; electrical burns; fire / property damage; regulatory fines	4 Likely	5 Catastrophic	20 VERY HIGH	No night work permitted without adequate lighting setup approved by HSE and confirmed in the Method Statement. No use of	Additional tower lights / portable floodlights for blind spots or high-risk zones. Daily photo records of lighting setup	1 Remote	5 Catastr ophic	5 MEDIUM	Electrical Engineer / Site Engineer / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		vehicles; poor visibility causing collision; electrical shock / electrocution from faulty lighting; trip hazard from trailing cables; glare / eye strain; fire from overloaded circuits					temporary lighting without Electrical PTW. Lighting levels maintained at a minimum of 200 lux in work areas, 50 lux in access routes (per DM CoCSP). LED floodlights / tower lights used to avoid glare. Cables routed safely, protected from damage and water. Overload protection devices installed on distribution boards. Electrical panels and generators placed in secure, bunded area away from water and traffic. Live cables clearly marked, secured, and covered with conduits or mats to prevent tripping. All temporary lighting installed and tested by competent certified electrician. Pre-use inspection and daily visual checks of lights, cables, and earthing. PTW system enforced. Emergency lighting (battery backup) provided in key areas. Pre-shift toolbox talk on night work hazards, visibility, and cable management. Mandatory high-visibility clothing safety helmet, safety boots, and eye protection (to reduce glare).	and lux level measurements. Immediate replacement / repair of any faulty light or cable. PPEs reinforced through daily pre-shift checks.				

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (H&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							All personnel inducted on night work lighting requirements and "no tampering" policy. Strict supervision by Electrical Engineer / HSE. Stop-work authority enforced for inadequate lighting or faults.					
32.	Demobilization	Equipment overturning during dismantling / loading; struck-by falling parts; rigging failure / load drop; collision with personnel / other plant; pinch points / crushing; unauthorized access to dismantling area; poor visibility during night demobilization; mechanical failure	Fatality; serious crush injury; permanent disability; major property damage; service disruption; regulatory fines	4 Likely	5 Catastrophic	20 VERY HIGH	Demobilization shall only commence after completion of piling work and confirmation from project management. Unnecessary personnel shall be cleared from dismantling area. No dismantling without approved procedure and PTW. Equipment dismantled following approved sequence and using certified lifting gear. Ground conditions verified before moving heavy machinery. Movement speed strictly limited to 20 km/h inside site. Work area fully barricaded with rigid barriers, cones, and warning signage. Safe routes identified and segregated for equipment removal. Exclusion zone maintained around dismantling and loading points. Activities supervised by Site Engineer and Piling Supervisor. Certified operators and riggers	Additional banksman assigned during equipment movement. Night demobilization only permitted with adequate lighting and extra supervision. Ground bearing pressure check and extra mats if soft ground. Daily photo records of dismantling progress and route condition. PPEs reinforced through pre-demobilization checks.	Remote	5 Catastrophic	5 MEDIUM	Construction Manager / Piling Supervisor / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HSAE)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							perform dismantling and loading. Pre-demobilization inspection of equipment and lifting gear. Communication is maintained via hand signals or radio. Emergency response plan activated. Mandatory safety helmet, high-visibility vest, safety shoes, gloves, and eye protection. Discipline: Toolbox talk conducted prior to demobilization explaining hazards and safe procedures. Strict supervision by Construction Manager / Piling Supervisor and HSE. Stop-work authority enforced for any unsafe condition.					
33.	Pile Testing (All Methods)	Structural instability and failure of reaction system during load testing Leading to falling equipment Uncontrolled load release Potential falls from test platform Confined space Electrical	Fatality Multiple serious crush injuries Burns Permanent disability Major property damage Failure of test pile Environmental contamination Regulatory penalties or work shutdown	4 Likely	5 Catastrophic	20 Very High	Pile testing shall not commence without an approved test procedure, valid Permit to Work (PTW), and formal approval from a competent Engineer Non-essential personnel are strictly excluded from the test zone Confined space entry is prohibited unless a full confined space permit is issued and rescue arrangements are verified All testing equipment – hydraulic jacks, load cells, reaction frames, kentledge blocks (mass certified), sonic sensors, CSL tubes, and TIP wires – is inspected,	Additional rigid fencing installs around test frame and kentledge stack to prevent unauthorized access Real-time data logging system is implemented with predefined alarm thresholds for displacement and load. immediate stop-work triggered for abnormal movement, signal loss or sensor malfunction Daily inspection records maintained for test setup,	1 Remote	5 Catastrophic	5 Medium	Testing Engineer / Construction Manager / HSE

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (H&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		Noise/vibration and spill hazards					calibrated/certified, and tagged before use Test load is applied gradually (maximum 25% increments) with continuous real-time monitoring of displacement, pressure, and settlement Sensors are lowered using mechanical winch only; manual handling prohibited Full exclusion zone (minimum 5m radius) established and barricaded with rigid barriers, cones, signage, and safety watchman Safety netting installed over pile heads; access tubes covered and locked when not in use Testing is supervised by qualified Testing Engineer with continuous monitoring of all parameters (load, displacement, settlement, sonic integrity, thermal data) Atmospheric testing (O ₂ >19.5%, LEL=0%, toxics safe) conducted before CSL/TIP confined space entry Emergency shutdown procedures and rescue plan established and briefed to all personnel Pre-test toolbox talk conducted covering hazards of all test	calibration and tube cleaning Additional spill kits provided for hydraulic oil or grout leakage Confined space rescue team and emergency equipment placed on standby during CSL/TIP operations PPE compliance reinforced through pre-test safety checks and supervision				

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS

DATE: 18-03-2026

SL NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							methods (static, dynamic, PIT, CSL, TIP) Mandatory PPE: safety helmet, high-visibility vest, safety footwear, gloves, hearing protection (>85 dBA), dust respirator, face shield (TIP hot wires), full body harness (height/confined space work) All personnel trained on method-specific hazards with strict supervision by Testing Engineer and HSE Officer Stop-work authority enforced for abnormal readings, equipment malfunction, or unsafe conditions					
34.	Use of Hand Tools	Cuts / lacerations from sharp edges; struck-by flying chips; repetitive strain / hand injury; improper tool storage causing trip hazard; misuse of chisels, hammers, crowbars during trimming / cleaning	Lacerations; amputation; musculoskeletal injury; eye injury	4 Likely	4 Major	16 VERY HIGH	No defective or damaged hand tools allowed on site. Only approved, inspected hand tools issued. Ergonomic grips and correct size tools used to reduce strain. Tool storage area clearly marked and segregated. Pre-use inspection by user. Toolbox talk on safe use and storage. Tools kept sharp and clean. Cut-resistant gloves, safety goggles, safety helmet, safety boots. Strict supervision by Supervisor. Defective tools immediately quarantined.	Daily tool issue / return log. Additional anti-vibration gloves for prolonged use.	1 Remote	4 Major	4 MEDIUM	Supervisor / Foreman / HSE
35.	Public Interface	Public intrusion into work zone; complaints / panic from noise /	Public injury; community complaint; regulatory penalty; project delay	3 Rare	5 CAT	15 HIGH	No public access allowed into site. Clear demarcation of site boundary. Advance notice to nearby residents / RTA. Noise & vibration monitoring	Additional public information boards with contact numbers. Joint interface meetings with RTA / DM.	1 Remote	5 CAT	5 Medium	Community Liaison / HSE / Constructi

HAZARD IDENTIFICATION & RISK ASSESSMENT

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (H&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
		vibration / dust; unauthorized photography / filming; pedestrian / vehicle conflict near site boundary; public exposure to hazards					with limits. Full perimeter hoarding / fencing with warning signs and CCTV. Community liaison officer appointed. Daily interface log maintained. Complaint response procedure in place. Strict security control. Any public approach immediately reported and escorted away.	Shadow vehicle for public safety during peak hours.				on Manager
36.	Environmental Controls (Dust, Noise & Vibration)	Dust generation from drilling / breaking; noise >85 dBA from rig / tools; vibration transmission to adjacent structures / utilities; air / water pollution from slurry / spoil	Respiratory illness; noise-induced hearing loss; structural damage to nearby buildings; environmental fines; community complaints	4 Likely	3 Moderate	12 High	No unnecessary idling of noisy equipment. Water suppression / misting for dust. Low-vibration tools and equipment maintenance. Noise barriers where required. Dust / noise / vibration zones demarcated with signage. Spoil stockpiles covered and away from water courses. Real-time monitoring (dust meters, noise dosimeters, vibration sensors). Daily records submitted to HSE. Complaint response procedure. Wastewater controlled via settling tanks. Dust masks / respirators, ear plugs / muffs, high-visibility clothing. Toolbox talks on environmental controls. Strict supervision and	Additional water tankers / misting cannons during dry / windy conditions. Vibration monitoring reports shared with RTA / DM. Extra noise barriers during night work.	1 Remote	3 Moderate	3 Low	HSE / Environmental Officer / Supervisor

HAZARD IDENTIFICATION & RISK ASSESSMENT

Reference Standard: TRA QHSS Management System Technical Guidelines (MSTG) - Section C.8

RISK ASSESSMENT FOR BORED CAST IN-SITU PILING WORKS											DATE: 18-03-2026	
SL. NO.	ACTIVITY	HAZARD / ASPECT	POTENTIAL IMPACT (HS&E)	INITIAL RISK RATING			EXISTING CONTROL MEASURES	ADDITIONAL CONTROL MEASURES	RESIDUAL RISK RATING			ACTION BY
				L	S	IRR			L	S	RRR	
							immediate corrective action on exceedance.					

Appendix B: PTW Procedure

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 35 of 39

PERMIT TO WORK (PTW) PROCEDURE

1. Purpose

The Permit to Work (PTW) system ensures that all high-risk activities are properly planned, authorized, controlled, and executed safely in compliance with project HSE requirements, regulatory standards (RTA/DM), and approved Method Statements.

2. Scope

This procedure applies to all high-risk activities including, but not limited to: Excavation, Lifting Operations, Work at Height, Hot Work, Electrical Work, Confined Space Entry, Earth Moving Equipment Operations, Pressure Testing, and any other activity identified as high-risk.

3. Roles & Responsibilities

Sl. No.	Role	Responsibility
1	Permit Receiver (Foreman / Supervisor - CCECC & Subcontractors)	Raise PTW, comply with all conditions, attend toolbox talk, stop work if unsafe, ensure site readiness
2	Permit Issuer (Engineer / Construction Manager)	Review documents, conduct site inspection, verify controls, issue PTW
3	Approver (CM / DPM / PM - Authorized by PD)	Review and provide final approval of PTW
4	Verifier (HSE Department)	Verify risk controls, ensure HSE compliance, issue unique PTW control number and maintain PTW Register

4. PTW Process

Sl. No.	Step	Description
1	Permit Request	Initiated by Foreman/Supervisor with attached approved Method Statement, Risk Assessment (MSTG C.8 compliant), drawings, and NOCs (if applicable)
2	Site Verification	Confirm safe access/egress, barricading & signage, utility identification & protection, availability of equipment and PPE
3	Risk Control Confirmation	Verify all control measures from the Risk Assessment are implemented, including emergency arrangements and competent personnel
4	Approval	Reviewed by Engineer and approved by CM / DPM / PM
5	PTW Numbering & Verification	HSE Department issues unique control number after full verification and records it in the PTW Register
6	Toolbox Talk / STARRT	Conduct briefing to explain hazards, control measures, and responsibilities; confirm workforce understanding
7	Work Execution	Work carried out strictly as per approved Method Statement and PTW conditions under continuous supervision
8	Monitoring & Control	HSE and supervision team perform regular inspections; stop work immediately if unsafe conditions are observed
9	Closure	Work area cleaned and restored, equipment removed safely, PTW signed off and closed

5. Validity & Control

Aspect	Requirement
Validity	Valid for specific activity, defined location, and limited time (typically one shift) except Excavation and Earth Moving Equipment permits will be issued for a week.
Renewal	Required if work extends beyond validity, scope changes, or site conditions change (e.g., weather).
Suspension	Immediate suspension in case of unsafe conditions, adverse weather, incident/near miss, or change in scope.
Emergency	Stop work immediately, inform Supervisor / HSE / Engineer, and follow the approved Project Emergency Preparedness and Response Plan (PEPRP).

6. Mandatory Requirement

No high-risk activity shall commence without a valid Permit to Work. Any violation will result in immediate work stoppage and corrective action.

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 36 of 39

Appendix C: Inspection & Test Plan

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 37 of 39

INSPECTION & TEST PLAN (ITP) FOR PILING WORKS

Client		M/s. Roads & Transport Authority (RTA)		Project	R1181-2 - IMPROVEMENT OF AL SAFA ROAD						
Consultant		M/s. Parsons		Discipline	Structures						
				ITP Ref. No.	CCECC-R1181-2-CC-ST-ITP-019				Rev	0	
Contractor		M/s. China Civil Engineering Construction Corporation – Dubai		Sub Contractor	MNG Zemtaz Soil & Foundation Engineering Construction Co. Inc				Date	08-Apr-26	
SN	Inspection / Test to be carried out	Frequency of Tests / Inspections	Acceptance Criteria	Applicable Specification / Standard	Control Data / Forms	Inspection Method	Responsibility				
							Subcontractor / Contractor	TPL	Consultant	Client / Authority	
1	Pre Approvals / Documentation										
1.1	NOC / Work permit from Relevant Authorities	Prior to start	Approved NOC Documents and comply all NOC conditions	Project Specification Vol-2 / Part 2 / Service works Clause 2/1 to 2/5	NOC Approval	Review	E	N/A	R	A	
1.2	Shop Drawing Approvals	Prior to start	Review and Approved by Engineer	Project Specification Vol-2 / Part 2 / Service works Clause 2/1 to 2/5	Submittal approval	Review	E	N/A	R/A	R/A	
1.3	Material Approvals	Prior to start	Review and approved by Engineer	Project Specification Vol-2 / Part 2 / Service works Clause 2/1 to 2/5	Submittal approval	Visual	E	N/A	W/A	R/A	
2	Construction Works										
2.1	Tool-Box Talk / Briefing on MSRA shall be carried out.	Prior to start	Ensure the Tool Box Talk conducted and documented.	Approved Method statements and Project Specification	Tool Box Briefing sheet	Review	E	N/A	R	R	
2.2	Survey Equipment Calibration	Prior to start	Third party certification (calibration of survey equipment by 3rd party). Valid calibration certificates for each equipment.	Project Specification	Calibration log with calibration certificates	Review	H	E	R	F	
2.3	Ensure the engineer informed via RFA before start the site activities	Prior to start	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	Work Notification	Work Notification	E	N/A	R/A	R	

INSPECTION & TEST PLAN (ITP) FOR PILING WORKS

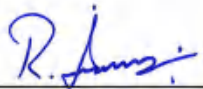
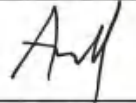
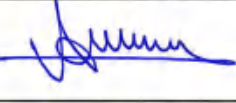
Client		M/s. Roads & Transport Authority (RTA)		Project	R1181-2 - IMPROVEMENT OF AL SAFA ROAD						
Consultant		M/s. Parsons		Discipline	Structures						
				ITP Ref. No.	CCECC-R1181-2-CC-ST-ITP-019				Rev	0	
Contractor		M/s. China Civil Engineering Construction Corporation – Dubai		Sub Contractor	MNG Zemas Soil & Foundation Engineering Construction Co. Inc				Date	08-Apr-26	
SN	Inspection / Test to be carried out	Frequency of Tests / Inspections	Acceptance Criteria	Applicable Specification / Standard	Control Data / Forms	Inspection Method	Responsibility				
							Subcontractor / Contractor	TPL	Consultant	Client / Authority	
2.4	Verification of existing services	Prior to start	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	NOC Approval	Visual	E	N/A	R	R	
2.5	Ensure the concerned Authority contacted to witness the work	Prior to start	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	NOC Approval	NOC	E	N/A	W	S	
2.6	Setting up Pile Location	Prior to start	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	RFA	Visual	E	N/A	W/A	R	
2.7	Temporary Guide Casing Fixing	Prior to start	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	RFA	Visual	E	N/A	R/A	S	
2.8	Drilling of Pile Bore Hole	Prior to start	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	RFA	Visual	E	N/A	R/A	R	
2.9	Steel Cage Installation	Prior to start	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	RFA	Visual	E	N/A	R/A	R	

INSPECTION & TEST PLAN (ITP) FOR PILING WORKS

Client	M/s. Roads & Transport Authority (RTA)	Project	R1181-2 - IMPROVEMENT OF AL SAFA ROAD			
Consultant	M/s. Parsons	Discipline	Structures			
		ITP Ref. No.	CCECC-R1181-2-CC-ST-ITP-019	Rev	0	
Contractor	M/s. China Civil Engineering Construction Corporation – Dubai	Sub Contractor	MNG Zentas Soil & Foundation Engineering Construction Co. Inc	Date	08-Apr-26	

SN	Inspection / Test to be carried out	Frequency of Tests / Inspections	Acceptance Criteria	Applicable Specification / Standard	Control Data / Forms	Inspection Method	Responsibility			
							Subcontractor / Contractor	TPL	Consultant	Client / Authority
2.10	Concrete Pouring	Prior to start	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	RFA	Visual	E	N/A	R/A	R
2.11	Final Inspection	Each location	Review and approved by Engineer	Approved Method Statements and Approved Shop Drawings & Project Specification	RFA	Visual	E	N/A	R/A	R

Note (1): H: Hold Point, W: Witness, R: Review, S: Surveillance, A: Approval, E : Execute, N/A : Not Applicable

	For Sub Contractor	For Contractor				For Consultant
Name	Srinivas	Raghavendra S	Syed Abdul Moid	Ahmed Abdalla	Ahmed Elkadi	
Designation	Technical Engineer	Project Engineer	QA/QC Manager	Technical Manager	Deputy Project Manager	
Signature						
Date	8-Apr-26	8-Apr-26	8-Apr-26	8-Apr-26	8-Apr-26	



Appendix D: Checklist

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 38 of 39

CHECKLIST FOR PILING WORKS

Project Name	: R1181-2 - Improvement of Al Safa Road	Report No.	:
Client	: Roads & Transport Authority	Pile Ref. ID	:
Consultant	: Parsons	Date of Boring	:
Contractor	: China Civil Engineering Construction Corporation – Dubai (CCECC)	Date of Casting	:
Piling Contractor	: M/s. MNG Zemas Soil & Foundation Eng Construction Co. Inc.		:

Item No.	Main Activity	Subcontractor	CCECC	Consultant	Remarks
1.0	General Requirements				
1.1	Start of Construction				
1.2	Technical Submittals (Method Statement, Shop Drawing etc.)				
1.3	Check the working platform area.				
1.4	Pre-check of all equipment & settings				
2.0	Casing Driving				
2.1	Check steel casing diameter, length, top & toe level				
2.2	Check for position deviation				
2.3	Allowable vertical deviation				
2.4	Check utilities before commencement of driving				
3.0	Pile Boring				
3.1	Ensure checking of borehole depth				
3.2	Ensure to check the borehole is properly cleaned prior to concreting				
4.0	Steel cage Installation				
4.1	Ensure steel cages fabricated in accordance to the approved construction drawing				
4.2	Ensure to check the borehole is properly cleaned prior to steel cage installation				
4.3	Ensure to check the top level of steel cage and sufficient concrete cover				
5.0	Concreting				
5.1	Ensure the Engineer's approval has been obtained prior to commencement of concrete pouring.				
5.2	Ensure the bottom of tremie pipe is not more than 0.30m from actual toe level of bore hole on the start of concreting of each pile.				
5.3	Ensure the minimum 2m embedment of tremie pipe to concrete.				
5.4	Check the elapse time of batching and placement of concrete did not exceed the maximum time requirements.				
5.5	Ensure the concrete is in accordance to the design mix.				
5.6	Ensure concrete temperature, slump of fresh concrete has been checked at all time of placement.				
5.7	Ensure concrete cube samples has been taken for testing.				

Sub.Contractor

CCECC

Consultant

Appendix E: Emergency Contact Details

Title:	Method Statement for Piling Works			R1181-2 - IMPROVEMENT OF AL SAFA ROAD	
Ref. No.:	CCECC-R1181-2-CC-ST-MSRA-019	Rev. No.:	01	Page No.:	Page 39 of 39

Emergency Contact Numbers

Contract No.:	R 1181-2	Project Title:	IMPROVEMENT OF AL SAFA STREET
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PROJECT DIRECTOR

Mohsen D Gazzaz	Mobile	052 282 5807
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DEPUTY PROJECT MANAGER

Ahmad Elkadi	Mobile	050 907 0342
Guo Pingcheng	Mobile	058 103 0678

HSE PERSONAL

Mirza Shahabuddin Baig	Mobile	055 933 1082
Ajmal Rashid	Mobile	056 148 7012
Shahid Ahmad	Mobile	050 560 9961
Basharat Karim	Mobile	054 358 9874
Asif Javed	Mobile	052 787 0311

CONSTRUCTION MANAGER

Ahmad Ghanem	Mobile	055 665 1144
Enji Shi	Mobile	054 738 1666

MALE NURSE

VIJAY SINGH	Mobile	058 837 2856
-------------	--------	--------------

SITE ENGINEER

Kasim (Utility & Road Workers)	Mobile	056 143 5466
Syed Abdul Kather (Electrical)	Mobile	055 789 3827
Hashim Ali (Structure)	Mobile	055 288 9793
Kishor Pote	Mobile	056 615 5815

SUPERVISOR

Zakir Ali	Mobile	055 168 6375
Gautam Kumar	Mobile	050 587 9443

EMERGENCY SERVICES

POLICE		999
CIVIL DEFENCE		997
AMBULANCE		998
DUBAI MUNICIPALITY		900
POISON CONTROL		800 424
RTA		800 9090
Iranian Hospital - Dubai	NEAREST HOSPITAL	04 344 0250

SITE ELECTRICIAN

Thomachan Thomas (Electrical Supervisor)	Mobile	055 280 4577
Sanjay	Mobile	052 368 7302